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1.2 Shell Scripts [f1ae60]

```
# ~/.bashrc
CXX="g++ -std=c++17 -Wall -Wextra"
DBG="-g -fsanitize=address,undefined -D_GLIBCXX_DEBUG"
test() { $CXX $DBG $1 && ./a.out "${@:2}"; }
run() { $CXX -O2 $1 && time ./a.out "${@:2}"; }
# ~/.vimrc
se nu ai et ru ic is sc cul re=1 ts=4 sts=4 sw=4 ls=2 mouse=a
```

1.3 Debug [33ccce]

```
# 對拍
CODE1="a"
CODE2="ac"
set -e
g++ $CODE1.cpp -o $CODE1
g++ $CODE2.cpp -o $CODE2
g++ gen.cpp -o gen
for ((i=1;;i++))
do
    echo "--- Testing: Case #${i} ---"
    ./gen > input
    # python3 gen.py > input
    ./CODE1 < input > $CODE1.out
    ./CODE2 < input > $CODE2.out
    cmp $CODE1.out $CODE2.out || break
done
# 多重解, ifstream in(argv[1]);
CODE="a"
set -e
g++ $CODE.cpp -o $CODE
g++ gen.cpp -o gen
g++ checker.cpp -o checker
for ((i=1;;i++))
do
    ./gen > input
    ./CODE < input > $CODE.out
    ./checker $CODE.out < input || break
done
# 互動
CODE="a"
set -e
g++ $CODE.cpp -o $CODE
g++ checker.cpp -o checker
PIPE_IN="in"
PIPE_OUT="out"
trap 'rm -f $PIPE_IN $PIPE_OUT' EXIT
mkfifo $PIPE_IN $PIPE_OUT
for ((i=1;;i++))
do
    echo "--- Testing: Case #${i} ---"
    ./CODE < $PIPE_IN > $PIPE_OUT &
    (exec 3>$PIPE_IN 4<$PIPE_OUT; ./checker <&4 >&3) || break
done
# 參考 checker
ll AC(int n) { return ans; }
void WA(string log = "") { cerr << log << endl; exit(1); }
void checkAC(string log = "") {
    string trash;
    if (cin >> trash) WA("redundant output\n" + log);
}
int main() {
    int n = uniform_int_distribution<int>(1, 10)(rng);
    ll sol = AC(n);
    stringstream log;
    cout << n << endl;
    log << n << endl;
    log << "judge: " << endl;
    log << "team: " << endl;
    WA(log.str());
    checkAC();
    return 0;
}
```

1.4 Compare Fuction [d41d8c]

```
// 1. sort, 二分搜刻在函式內 lambda 就好
// 2. priority queue 小到是大 >, set 是 <
// 3. set 不能 =, multiset 必須 =
// 4. 確保每個成員都要比到
// 5. pbds_multiset 不要用 lower_bound
// 6. 如果要用 find, 插入 inf 後使用 upper_bound
// 7. multiset 可以跟 set 一樣使用, 但請注意第 3、4 點
auto cmp = [](int i, int j) { return i > j; };
priority_queue<int, vector<int>, decltype(cmp)> pq(cmp);

vector<int> a {1, 2, 5, 4, 3}; // 小心不要改到 a
auto cmp = [&a](int i, int j) { return a[i] > a[j]; };
priority_queue<int, vector<int>, decltype(cmp)> pq(cmp);

vector<int> v {1, 2, 3, 4, 5};
upper_bound(v.begin(), v.end(), 2, [](int a, int b)
{ return a < b; }); // find first b that a < b, a is 2
lower_bound(v.begin(), v.end(), 2, [](int a, int b)
{ return a < b; }); // find first a that a < b fail, b is 2
```

1.5 Pbds [d41d8c]

```
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
```

1 Basic

1.1 Default Code [d41d8c]

```
#include <bits/stdc++.h>

using namespace std;
using ll = long long;

namespace rgs = ranges;
mt19937 rng
    (chrono::steady_clock::now().time_since_epoch().count());

template<class T> bool
chmin(T &a, const T &b) { return a > b && (a = b, true); }
template<class T> bool
chmax(T &a, const T &b) { return a < b && (a = b, true); }

void solve() {
}

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);

    int t = 1;
    cin >> t;
    while (t--) {
        solve();
    }

    return 0;
}
```

```
using namespace __gnu_pbds;
template<class T, class F = less<T>>
using pbds_set = tree<T, null_type
, F, rb_tree_tag, tree_order_statistics_node_update>;
template<class T, class F = less_equal<T>>
using pbds_multiset = tree<T, null_type
, F, rb_tree_tag, tree_order_statistics_node_update>;
```

1.6 Int128 [eb8585]

```
using i128 = __int128_t; // 1.7E38
istream &operator>>(istream &is, i128 &a) {
    i128 sgn = 1; a = 0;
    string s; is >> s;
    for (auto c : s) {
        if (c == '-') sgn = -1;
        else a = a * 10 + c - '0';
    }
    a *= sgn;
    return is;
}
ostream &operator<<(ostream &os, i128 a) {
    if (a == 0) return os << 0;
    string res;
    if (a < 0) os << '-', a = -a;
    while (a) res.push_back(a % 10 + '0'), a /= 10;
    reverse(res.begin(), res.end());
    os << res;
    return os;
}
```

1.7 Bitset 01 String [cc11a2]

```
#pragma GCC target("popcnt")
bitset<60> b = x; // 讀整數轉 bitset
cin >> b; // 讀 01 字串轉 bitset
b.to_ullong(); // bitset 轉整數
b.to_string(); // bitset 轉 01 字串
stoll(s, 0, 2); // 01 字串轉成 ll
```

2 Graph

2.1 Binary Lifting LCA [4399db]

```
const int N = 2E5;
const int Lg = __lg(N); // __lg(max(N, Qi)), [0, Lg]
int up[N][Lg + 1];
vector<int> dep, dfn;
void build(int n, vector<vector<int>> &g, int rt = 0) {
    dep.assign(n, 0); dfn.assign(n, 0);
    int cur = 0;
    [&](this auto &&self, int x, int p) -> void {
        dfn[x] = cur++;
        up[x][0] = p;
        for (int i = 1; i <= Lg; i++) {
            int nxt = up[x][i - 1];
            up[x][i] = up[nxt][i - 1];
        }
        for (auto y : g[x]) {
            if (y == p) continue;
            up[y][0] = x;
            dep[y] = dep[x] + 1;
            self(y, x);
        }
    }(rt, rt);
}
int lca(int a, int b) {
    if (dep[a] < dep[b]) swap(a, b);
    int pull = dep[a] - dep[b];
    for (int i = 0; i <= Lg; i++)
        if (pull >> i & 1) a = up[a][i];
    if (a == b) return a;
    for (int i = Lg; i >= 0; i--)
        if (up[a][i] != up[b][i])
            a = up[a][i], b = up[b][i];
    return up[a][0];
}
int jump(int x, int k) {
    for (int i = Lg; i >= 0; i--)
        if (k >> i & 1) x = up[x][i];
    return x;
}
int dist(int a, int b) {
    return dep[a] + dep[b] - 2 * dep[lca(a, b)];
}
```

2.2 Heavy Light Decomposition [dc98bc]

```
struct HLD {
    int n, cur = 0;
    vector<int> siz, top, dep, par, in, out, seq;
    vector<vector<int>> adj;
    HLD(int n) : n(n) {
        siz.resize(n); top.resize(n); dep.resize(n);
        par.resize(n); in.resize(n); out.resize(n);
        seq.resize(n); adj.assign(n, {});
    }
    void addEdge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }
}
```

```
}
void work(int rt = 0) {
    top[rt] = rt;
    dep[rt] = 0;
    par[rt] = -1;
    dfs1(rt); dfs2(rt);
}
void dfs1(int u) {
    if (par[u] != -1) adj[u].erase(find(adj[u].begin(), adj[u].end(), par[u]));
    siz[u] = 1;
    for (auto &v : adj[u]) {
        par[v] = u, dep[v] = dep[u] + 1;
        dfs1(v);
        siz[u] += siz[v];
        if (siz[v] > siz[adj[u][0]]) {
            swap(v, adj[u][0]);
        } // 讓 adj[u][0] 是重子節點
    }
}
void dfs2(int u) {
    in[u] = cur++;
    seq[in[u]] = u; // dfn 對應的編號
    for (auto v : adj[u]) {
        top[v] = v == adj[u][0] ? top[u] : v;
        dfs2(v);
    }
    out[u] = cur;
}
int lca(int u, int v) {
    while (top[u] != top[v]) {
        if (dep[top[u]] > dep[top[v]]) {
            u = par[top[u]];
        } else {
            v = par[top[v]];
        }
    }
    return dep[u] < dep[v] ? u : v;
}
int dist(int u, int v) {
    return dep[u] + dep[v] - 2 * dep[lca(u, v)];
}
int jump(int u, int k) {
    if (dep[u] < k) return -1;
    int d = dep[u] - k;
    while (dep[top[u]] > d) u = par[top[u]];
    return seq[in[u] - dep[u] + d];
}
bool isAncestor(int u, int v) {
    return in[u] <= in[v] && in[v] < out[u];
}
int rootedParent(int rt, int v) {
    if (rt == v) return rt;
    if (!isAncestor(v, rt)) return par[v];
    auto it = upper_bound(adj[v].begin(), adj[v].end(), rt,
        [&](int x, int y) {
            return in[x] < in[y];
        }) - 1;
    return *it;
}
int rootedSize(int rt, int v) {
    if (rt == v) return n;
    if (!isAncestor(v, rt)) return siz[v];
    return n - siz[rootedParent(rt, v)];
}
int rootedLca(int rt, int a, int b) {
    return lca(rt, a) ^ lca(a, b) ^ lca(b, rt);
}
}
```

2.3 Centroid Decomposition [7ef37f]

```
vector<bool> vis(n);
vector<int> siz(n), par(n, -1);
auto findSize = [&](auto &&self, int u, int p) -> int {
    siz[u] = 1;
    for (int v : g[u]) {
        if (v == p || vis[v]) continue;
        siz[u] += self(self, v, u);
    }
    return siz[u];
};
auto findCen = [&](auto &&self, int u, int p, int sz) -> int {
    for (int v : g[u]) {
        if (v == p || vis[v]) continue;
        if (siz[v] * 2 > sz) return self(self, v, u, sz);
    }
    return u;
};
auto build = [&](auto &&self, int u, int p) -> void {
    findSize(findSize, u, p);
    int c = findCen(findCen, u, -1, siz[u]);
    vis[c] = true, par[c] = p;
    for (int v : g[c]) if (!vis[v]) self(self, v, c);
}; build(build, 0, -1);
```

2.4 SCC [fae635]

```
struct SCC {
    int n, cur = 0, cnt = 0;
    vector<vector<int>> adj;
    vector<int> stk, dfn, low, bel;
```

```

SCC(int n) : n(n), adj(n), dfn(n, -1), low(n), bel(n, -1) {}
void addEdge(int u, int v) { adj[u].push_back(v); }
void dfs(int x) {
    dfn[x] = low[x] = cur++;
    stk.push_back(x);
    for (auto y : adj[x]) {
        if (dfn[y] == -1) {
            dfs(y);
            low[x] = min(low[x], low[y]);
        } else if (bel[y] == -1) {
            low[x] = min(low[x], dfn[y]);
        }
    }
    if (dfn[x] == low[x]) {
        int y;
        do {
            y = stk.back();
            bel[y] = cnt;
            stk.pop_back();
        } while (y != x);
        cnt++;
    }
}
vector<int> work() {
    for (int i = 0; i < n; i++)
        if (dfn[i] == -1) dfs(i);
    return bel;
}
struct Graph {
    int n;
    vector<pair<int, int>> edges;
    vector<int> siz, cnte;
};
Graph compress() {
    Graph g; g.n = cnt;
    g.siz.resize(cnt); g.cnte.resize(cnt);
    for (int i = 0; i < n; i++) {
        g.siz[bel[i]]++;
        for (auto j : adj[i]) {
            if (bel[i] != bel[j]) {
                g.edges.emplace_back(bel[i], bel[j]);
            } else {
                g.cnte[bel[i]]++;
            }
        }
    }
    return g;
};

```

2.5 2-SAT [093413]

```

struct TwoSat : SCC {
    vector<int> ans;
    TwoSat(int n) : SCC(2 * n), ans(n) {}
    void addClause(int u, bool a, int v, bool b) {
        addEdge(2 * u + !a, 2 * v + b);
        addEdge(2 * v + !b, 2 * u + a);
    }
    void ifThen(int u, bool a, int v, bool b) {
        // 必取 A: not A -> A
        addEdge(2 * u + a, 2 * v + b);
    }
    bool work() {
        SCC::work(); // bel[!a] > bel[a] => !a -> a
        for (int i = 0; i < n / 2; i++) {
            if (bel[2 * i] == bel[2 * i + 1]) return false;
            ans[i] = bel[2 * i] > bel[2 * i + 1];
        }
        return true;
    }
};

```

2.6 EBCC [a84a20]

```

struct EBCC { // CF/contest/1986/pF
    int n, cur = 0, cnt = 0;
    vector<vector<int>> adj;
    vector<int> stk, dfn, low, bel;
    vector<pair<int, int>> bridges; // 關鍵邊
    EBCC(int n) : n(n), adj(n), low(n), dfn(n, -1), bel(n, -1) {}
    void addEdge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }
    void dfs(int x, int p) {
        dfn[x] = low[x] = cur++;
        stk.push_back(x);
        for (auto y : adj[x]) {
            if (y == p) continue;
            if (dfn[y] == -1) {
                dfs(y, x);
                low[x] = min(low[x], low[y]);
                if (low[y] > dfn[x]) {
                    bridges.emplace_back(x, y);
                }
            } else if (bel[y] == -1) {
                low[x] = min(low[x], dfn[y]);
            }
        }
        if (dfn[x] == low[x]) {
            int y;

```

```

            do {
                y = stk.back();
                bel[y] = cnt;
                stk.pop_back();
            } while (y != x);
            cnt++;
        }
    }
    vector<int> work() { // not connected
        for (int i = 0; i < n; i++)
            if (dfn[i] == -1) dfs(i, -1);
        return bel;
    }
    struct Graph {
        int n;
        vector<pair<int, int>> edges;
        vector<int> siz, cnte;
    };
    Graph compress() {
        Graph g; g.n = cnt;
        g.siz.resize(cnt); g.cnte.resize(cnt);
        for (int i = 0; i < n; i++) {
            g.siz[bel[i]]++;
            for (auto j : adj[i]) {
                if (bel[i] < bel[j]) {
                    g.edges.emplace_back(bel[i], bel[j]);
                } else if (bel[i] == bel[j] && i < j) {
                    g.cnte[bel[i]]++;
                }
            }
        }
        return g;
    }
};

```

2.7 VBCC [775dfa]

```

struct VBCC {
    int n, cur = 0, cnt = 0;
    vector<vector<int>> adj, bcc;
    vector<int> dfn, low, stk;
    vector<bool> ap;
    VBCC(int n) : n(n), adj(n), dfn(n, 0), low(n, 0), ap(n) {}
    void addEdge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }
    void dfs(int x, int p = -1) {
        int ch = 0;
        dfn[x] = low[x] = ++cur;
        stk.push_back(x);
        for (int y : adj[x]) {
            if (!dfn[y]) {
                dfs(y, x);
                ch++;
                low[x] = min(low[x], low[y]);
                if (dfn[x] <= low[y]) {
                    ap[x] = true;
                    bcc.emplace_back();
                    int v;
                    do {
                        v = stk.back();
                        stk.pop_back();
                        bcc.back().push_back(v);
                    } while (v != y);
                    bcc.back().push_back(x);
                    cnt++;
                }
            } else if (dfn[y] < dfn[x] && y != p) {
                low[x] = min(low[x], dfn[y]);
            }
        }
        if (p == -1 && ch < 2) ap[x] = false;
    }
    void work() {
        for (int i = 0; i < n; i++)
            if (!dfn[i])
                if (adj[i].empty()) {
                    bcc.emplace_back(1, i);
                    cnt++;
                } else dfs(i);
    }
    struct Graph {
        int n;
        vector<pair<int, int>> edges;
        vector<int> bel, siz, cnte;
    };
    Graph compress() {
        Graph g;
        g.bel.assign(n, -1);
        g.siz.assign(cnt, 0);
        g.cnte.assign(cnt, 0);
        g.n = cnt;
        for (int i = 0; i < n; i++) {
            if (ap[i]) {
                g.bel[i] = g.n++;
                g.siz.push_back(1);
                g.cnte.push_back(0);
            }
        }
        vector<bool> in_bcc(n);
        for (int i = 0; i < cnt; i++) {
            for (int u : bcc[i]) in_bcc[u] = true;

```

```

int edges = 0;
for (int u : bcc[i]) {
    if (ap[u]) g.edges.emplace_back(g.bel[u], i);
    else g.bel[u] = i, g.sz[i]++;
    for (int v : adj[u]) if (in_bcc[v]) edges++;
}
g.cnte[i] = edges / 2;
for (int u : bcc[i]) in_bcc[u] = false;
}
return g;
};

```

2.8 Functional Graph [060d10]

```

const int N = 2E5;
const int Lg = __lg(N); // __lg(max(n, qi)), [0, Lg]
int cht[N][Lg];
struct FunctionalGraph {
    int n, cnt = 0;
    vector<int> g, bel, id, cycsz, in, top, hei;
    FunctionalGraph(const vector<int> &g) : n(g.size()) {
        g(g), bel(n, -1), id(n), in(n), top(n, -1), hei(n) {
            for (int i = 0; i < n; i++)
                cht[i][0] = g[i], in[g[i]]++;
            for (int i = 1; i <= Lg; i++)
                for (int u = 0; u < n; u++) {
                    int nxt = cht[u][i - 1];
                    cht[u][i] = cht[nxt][i - 1];
                }
            for (int i = 0; i < n; i++)
                if (in[i] == 0) label(i);
            for (int i = 0; i < n; i++)
                if (top[i] == -1) label(i);
        }
    }
    void label(int u) {
        vector<int> p; int cur = u;
        while (top[cur] == -1) {
            top[cur] = u;
            p.push_back(cur);
            cur = g[cur];
        }
        auto s = find(p.begin(), p.end(), cur);
        vector<int> cyc(s, p.end());
        p.erase(s, p.end()); p.push_back(cur);
        for (int i = 0; i < (int)cyc.size(); i++)
            bel[cyc[i]] = cnt, id[cyc[i]] = i, hei[cyc[i]] = cyc.size();
        if (!cyc.empty())
            ++cnt, cycsz.push_back(cyc.size());
        for (int i = p.size() - 1; i > 0; i--)
            id[p[i] - 1] = id[p[i]] - 1, hei[p[i] - 1] = hei[p[i]] + 1;
    }
    int jump(int u, int k) {
        for (int b = 0; k > 0; b++) {
            if (k & 1) u = cht[u][b];
            k >>= 1;
        }
        return u;
    }
};

```

2.9 Bellman-Ford [430de2]

```

// 用 Bellman Ford 找負環
void bellmanFord() {
    int n, m; cin >> n >> m;
    vector<array<int, 3>> e;
    for (int i = 0; i < m; i++) {
        int u, v, w; cin >> u >> v >> w;
        u--, v--; e.push_back({u, v, w});
    }
    vector<ll> dis(n, inf), par(n);
    int t = -1; dis[0] = 0;
    for (int i = 1; i <= n; i++) {
        for (auto [u, v, w] : e) {
            if (dis[v] > dis[u] + w) {
                dis[v] = dis[u] + w;
                par[v] = u;
                if (i == n) t = v;
            }
        }
    }
    if (t == -1) { cout << "NO\n"; return; }
    for (int i = 1; i < n; i++) t = par[t];
    vector<int> ans {t};
    int i = t;
    do {
        i = par[i];
        ans.push_back(i);
    } while (i != t);
    reverse(ans.begin(), ans.end());
    cout << "YES\n";
    for (auto x : ans) cout << x + 1 << " ";
}

```

2.10 Euler [13ce48]

```

// 1. 無向圖是歐拉圖：
// 非零度頂點是連通的

```

```

// 頂點的度數都是偶數

```

```

// 2. 無向圖是半歐拉圖(有路沒有環)：
// 非零度頂點是連通的
// 恰有 2 個奇度頂點

```

```

// 3. 有向圖是歐拉圖：
// 非零度頂點是強連通的
// 每個頂點的入度和出度相等

```

```

// 4. 有向圖是半歐拉圖(有路沒有環)：
// 非零度頂點是弱連通的
// 至多一個頂點的出度與入度之差為 1
// 至多一個頂點的入度與出度之差為 1
// 其他頂點的入度和出度相等

```

```

vector<int> ans;
[&](this auto &&self, int u) -> void {
    while (g[u].size()) {
        int v = *g[u].begin();
        g[u].erase(v);
        self(v);
    }
    ans.push_back(u);
} (0); reverse(ans.begin(), ans.end());

```

2.11 Virtual Tree [b2b0eb]

```

// 多次詢問給某些關鍵點，虛樹可達成快速樹 DP (前處理每個點)
// 例如這題是有權樹，給一些關鍵點，求跟 vertex 1 隔開的最小成本
// 前處理 root 到所有點的最小邊權

```

```

void solve(int n, vector<vector<pair<int, int>>> &g, int q) {
    build(n, g); // build LCA
    vector<int> dis(n, 1E9); // root 到各點的最小邊權
    [&](this auto &&self, int x, int p) -> void {
        for (auto [y, w] : g[x]) {
            if (y == p) continue;
            dis[y] = min(w, dis[x]);
            self(y, x);
        }
    } (0, -1);
    vector<vector<int>> vt(n);
    vector<int> isKey(n);
    vector<ll> dp(n);
    while (q--) {
        int m; cin >> m;
        vector<int> key(m);
        for (int i = 0; i < m; i++) {
            cin >> key[i];
            isKey[--key[i]] = 1;
        }
        key.push_back(0); // 固定 0 為 root，看題目需求
        sort(key.begin(), key.end(), [&](int a, int b) {
            return dfn[a] < dfn[b];
        }); // 要 sort 再 insert
        vector<int> stk; // 維護一條虛樹鏈
        for (int k : key) {
            if (stk.empty()) {
                stk.push_back(k);
                continue;
            }
            int l = lca(stk.back(), k);
            if (l == stk.back()) { // 直接一條鏈
                stk.push_back(k);
                continue;
            } // 次頂是 lca 的子孫，直接跟最頂連邊
            while (stk.size() > 1 && dfn[stk[stk.size() - 2]] > dfn[l]) {
                vt[stk[stk.size() - 2]].push_back(stk.back());
                stk.pop_back();
            } // 保證 back 是 lca 的子孫，且次頂不是 lca 的子孫
            vt[l].push_back(stk.back());
            if (stk.size() < 2 || stk[stk.size() - 2] != l) stk.back() = l;
            else stk.pop_back(); // lca 的子孫已經用不到了，直接換成 lca
            stk.push_back(k);
        }
        while (stk.size() > 1) {
            vt[stk[stk.size() - 2]].push_back(stk.back());
            stk.pop_back();
        }
        int rt = stk[0]; stk.clear();
        [&](this auto &&self, int x) -> void {
            for (auto y : vt[x]) {
                self(y);
                if (isKey[y]) { // 直接砍了
                    dp[x] += dis[y];
                } else { // 不砍 or 砍
                    dp[x] += min<ll>(dp[y], dis[y]);
                } // 記得 reset
                isKey[y] = dp[y] = 0;
            }
            vt[x].clear(); // 記得 reset
        } (rt);
        cout << dp[rt] << "\n";
        isKey[rt] = 0;
        dp[rt] = 0; // 最後 reset root
    }
}

```

```
}
}
```

2.12 Dominator Tree [a54c17]

```
// dom
存起點到達此點的必經的上個節點(起點 = 自己), 無法到達 = -1
struct DominatorTree {
    int n, id = 0;
    vector<vector<int>> adj, radj, bucket;
    vector<int> sdom, dom, vis, rev, pa, rt, mn, res;
    DominatorTree(int
        n) : n(n), adj(n), radj(n), bucket(n), sdom(n), dom(n),
        -1), vis(n, -1), rev(n), pa(n), rt(n), mn(n), res(n) {}
    void addEdge(int u, int v) { adj[u].push_back(v); }
    int query(int v, int x) {
        if (rt[v] == v) return x ? -1 : v;
        int p = query(rt[v], 1);
        if (p == -1) return x ? rt[v] : mn[v];
        if (sdom[mn[v]] > sdom[mn[rt[v]]]) mn[v] = mn[rt[v]];
        rt[v] = p;
        return x ? p : mn[v];
    }
    void dfs(int v) {
        vis[v] = id, rev[id] = v;
        rt[id] = mn[id] = sdom[id] = id, id++;
        for (int u : adj[v]) {
            if (vis[u] == -1) dfs(u), pa[vis[u]] = vis[v];
            radj[vis[u]].push_back(vis[v]);
        }
    }
    vector<int> build(int s) {
        dfs(s);
        for (int i = id - 1; i >= 0; i--) {
            for (int u : radj[i])
                sdom[i] = min(sdom[i], sdom[query(u, 0)]);
            if (i) bucket[sdom[i]].push_back(i);
            for (int u : bucket[i]) {
                int p = query(u, 0);
                dom[u] = sdom[p] == i ? i : p;
            }
            if (i) rt[i] = pa[i];
        }
        res.assign(n, -1);
        for (int i = 1; i < id; i++)
            if (dom[i] != sdom[i]) dom[i] = dom[dom[i]];
        for (int i = 1; i < id; i++) res[rev[i]] = rev[dom[i]];
        res[s] = s;
        for (int i = 0; i < n; i++) dom[i] = res[i];
        return dom;
    }
};
```

3 Data Structure

3.1 DSU [d41d8c]

```
struct DSU {
    int n;
    vector<int> f, siz;
    DSU(int n) : n(n), f(n), siz(n, 1)
    { iota(f.begin(), f.end(), 0); }
    int find(int x) { return x == f[x] ? x : f[x] = find(f[x]); }
    bool same(int x, int y) { return find(x) == find(y); }
    bool merge(int x, int y) {
        x = find(x), y = find(y);
        if (x == y) return false;
        if (siz[x] < siz[y]) swap(x, y);
        siz[x] += siz[y];
        f[y] = x;
        n--;
        return true;
    }
    int size(int x) { return siz[find(x)]; }
};

struct DSU {
    int n;
    vector<int> f, siz, stk;
    DSU(int n) : n(n), f(n), siz(n, 1)
    { iota(f.begin(), f.end(), 0); }
    int find(int x) { return x == f[x] ? x : find(f[x]); }
    bool same(int x, int y) { return find(x) == find(y); }
    bool merge(int x, int y) {
        x = find(x), y = find(y);
        if (x == y) return false;
        if (siz[x] < siz[y]) swap(x, y);
        siz[x] += siz[y];
        f[y] = x;
        n--;
        stk.push_back(y);
        return true;
    }
    void undo(int x) {
        while (stk.size() > x) {
            int y = stk.back();
            stk.pop_back();
            n++;
            siz[f[y]] -= siz[y];
            f[y] = y;
        }
    }
    int size(int x) { return siz[find(x)]; }
};
```

3.2 Fenwick [d41d8c]

```
template<class T> struct Fenwick {
    int n; vector<T> a;
    Fenwick(int n) : n(n), a(n) {}
    void add(int x, const T &v) {
        for (int i = x + 1; i <= n; i += i & -i)
            a[i - 1] = a[i - 1] + v;
    }
    T sum(int x) {
        T ans{};
        for (int i = x; i > 0; i -= i & -i)
            ans = ans + a[i - 1];
        return ans;
    }
    T rangeSum(int l, int r) { return sum(r) - sum(l); }
    int select(const T &k, int start = 0) {
        // 找到最小的 x, 使得 sum(x + 1) - sum(start) > k
        // prefix sum 要有單調性
        int x = 0; T cur = -sum(start);
        for (int i = 1 << __lg(n); i; i /= 2)
            if (x + i <= n && cur + a[x + i - 1] <= k)
                x += i, cur = cur + a[x - 1];
        return x;
    }
};

template<class T> struct Fenwick2D {
    int n, m;
    vector<vector<T>> a;
    Fenwick2D(int n, int m) : n(n), m(m), a(n, vector<T>(m)) {}
    void add(int x, int y, const T &v) {
        for (int i = x + 1; i <= n; i += i & -i)
            for (int j = y + 1; j <= m; j += j & -j)
                a[i - 1][j - 1] = a[i - 1][j - 1] + v;
    }
    T sum(int x, int y) {
        T ans{};
        for (int i = x; i > 0; i -= i & -i)
            for (int j = y; j > 0; j -= j & -j)
                ans = ans + a[i - 1][j - 1];
        return ans;
    }
    T rangeSum(int x1, int y1, int x2, int y2) {
        return
            sum(x2, y2) - sum(x1, y2) - sum(x2, y1) + sum(x1, y1);
    }
};
```

3.3 Range Fenwick [d41d8c]

```
template<class T> struct RangeFenwick {
    int n;
    vector<T> d, di;
    RangeFenwick(int n) : n(n), d(n), di(n) {}
    void add(int x, const T &v) {
        T vi = v * (x + 1);
        for (int i = x + 1; i <= n; i += i & -i)
            d[i - 1] = d[i - 1] + v, di[i - 1] = di[i - 1] + vi;
    }
    void rangeAdd
        (int l, int r, const T &v) { add(l, v), add(r, -v); }
    T sum(int x) {
        T ans{};
        for (int i = x; i > 0; i -= i & -i)
            ans = ans + T(x + 1) * d[i - 1] - di[i - 1];
        return ans;
    }
    T rangeSum(int l, int r) { return sum(r) - sum(l); }
    int select(const T &k, int start = 0) {
        int x = 0;
        T cur {}, curi {}, sub = sum(start);
        for (int i = 1 << __lg(n); i; i /= 2)
            if (x + i <= n && T(x + i + 1) * (cur +
                d[x + i - 1]) - (curi + di[x + i - 1]) - sub <= k) {
                x += i;
                cur = cur + d[x - 1], curi = curi + di[x - 1];
            }
        return x;
    }
};

template<class T> struct RangeFenwick2D {
    int n, m;
    vector<vector<T>> d, di, dj, dij;
    RangeFenwick2D
        (int n, int m) : n(n), m(m), dij(n, vector<T>(m))
    { d = di = dj = dij; }
    void add(int x, int y, const T &v) {
        T vi = v * (x + 1), vj = v * (y + 1);
        T vij = v * (x + 1) * (y + 1);
        for (int i = x + 1; i <= n; i += i & -i)
            for (int j = y + 1; j <= m; j += j & -j) {
                d[i - 1][j - 1] = d[i - 1][j - 1] + v;
                di[i - 1][j - 1] = di[i - 1][j - 1] + vi;
                dj[i - 1][j - 1] = dj[i - 1][j - 1] + vj;
                dij[i - 1][j - 1] = dij[i - 1][j - 1] + vij;
            }
    }
    void rangeAdd(int x1, int y1, int x2, int y2, const T &v) {
        add(x1, y1, v), add(x2, y2, v);
        add(x1, y2, -v), add(x2, y1, -v);
    }
    T sum(int x, int y) {
```

```

T ans{};
for (int i = x; i > 0; i -= i & -i)
    for (int j = y; j > 0; j -= j & -j)
        ans = ans + (x + 1) * (y + 1) * d[i - 1][j - 1]
            - (y + 1) * d[i - 1][j]
            - (x + 1) * d[j - 1][i - 1]
            + d[j - 1][i];
return ans;
}
T rangeSum(int x1, int y1, int x2, int y2)
{
    return
        sum(x2, y2) - sum(x1, y2) - sum(x2, y1) + sum(x1, y1);
};

```

3.4 Zkw Segment Tree [d41d8c]

```

template<class T> struct ZKW {
    int n;
    vector<T> a;
    ZKW(int n) : n(n), a(2 * n) {}
    ZKW(const vector<T> &v) : n(v.size()), a(2 * n) {
        for (int i = 0; i < n; i++) a[i + n] = v[i];
        for (int i = n - 1; i > 0; i--) a[i] = a[2 * i] + a[2 * i + 1];
    }
    void modify(int x, const T &i) {
        a[x + n] = i;
        for (x /= 2; x; x /= 2) a[x] = a[2 * x] + a[2 * x + 1];
    }
    T query(int l, int r) {
        T x{}, y{};
        for (l += n, r += n; l < r; l /= 2, r /= 2) {
            if (l & 1) x = x + a[l++];
            if (r & 1) y = a[--r] + y;
        }
        return x + y;
    }
};

```

3.5 Lazy Segment Tree [d41d8c]

```

template<class Info, class Tag> struct LazySegmentTree {
    int n;
    vector<Info> info;
    vector<Tag> tag;
    LazySegmentTree(
        int n) : n(n), info(4 << __lg(n)), tag(4 << __lg(n)) {}
    LazySegmentTree(const vector<Info> &v) : n(v.size()), info(4 << __lg(n)), tag(4 << __lg(n)) {}
    auto build = [&](auto &&self, int l, int r, int p) {
        if (r - l == 1) {
            info[p] = v[l];
            return;
        }
        int m = (l + r) / 2;
        self(self, l, m, 2 * p);
        self(self, m, r, 2 * p + 1);
        pull(p);
    };
    build(build, 0, n, 1);
    void pull(int p) { info[p] = info[2 * p] + info[2 * p + 1]; }
    void apply(const Tag &t, int l, int r, int p) {
        info[p].apply(t, l, r);
        tag[p].apply(t);
    }
    void push(int l, int r, int p) {
        int m = (l + r) / 2;
        apply(tag[p], l, m, 2 * p);
        apply(tag[p], m, r, 2 * p + 1);
        offset(m - l, m, r, 2 * p + 1); // tag 可位移 (m - l)
        tag[p] = Tag();
    }
    void modify(int x, const Info &i) { modify(x, i, 0, n, 1); }
    void modify(int x, const Info &i, int l, int r, int p) {
        if (r - l == 1) {
            info[p] = i;
            return;
        }
        int m = (l + r) / 2;
        push(l, r, p);
        if (x < m) modify(x, i, l, m, 2 * p);
        else modify(x, i, m, r, 2 * p + 1);
        pull(p);
    }
    Info query(int ql, int qr) { return query(ql, qr, 0, n, 1); }
    Info query(int ql, int qr, int l, int r, int p) {
        if (qr <= l || ql >= r) return Info();
        if (ql <= l && r <= qr) return info[p];
        int m = (l + r) / 2;
        push(l, r, p);
        return query(ql, qr, l, m, 2 * p) + query(ql, qr, m, r, 2 * p + 1);
    }
    void rangeApply(int ql, int qr, const Tag &t) { rangeApply(ql, qr, t, 0, n, 1); }
    void rangeApply(int ql, int qr, const Tag &t, int l, int r, int p) {
        if (qr <= l || ql >= r) return;
        if (ql <= l && r <= qr) {
            apply(t.offset(l - ql), l, r, p); // tag 可位移 (l - ql)
            return;
        }
        int m = (l + r) / 2;
        push(l, r, p);
        rangeApply(ql, qr, t, l, m, 2 * p);
        rangeApply(ql, qr, t, m, r, 2 * p + 1);
        pull(p);
    }
};

```

```

}
int m = (l + r) / 2;
push(l, r, p);
rangeApply(ql, qr, t, l, m, 2 * p);
rangeApply(ql, qr, t, m, r, 2 * p + 1);
pull(p);
}
template<class F> int findFirst(int ql, int qr, F &&f) {
    return findFirst(ql, qr, f, 0, n, 1);
} // 若要找 last, 先右子樹遞迴即可
template<class F> int findFirst(int ql, int qr, F &&f, int l, int r, int p) {
    if (qr <= l || ql >= r) return -1;
    if (ql <= l && r <= qr && !f(info[p])) return -1;
    if (r - l == 1) return l;
    int m = (l + r) / 2;
    push(l, r, p);
    int res = findFirst(ql, qr, f, l, m, 2 * p);
    if (res == -1) res = findFirst(ql, qr, f, m, r, 2 * p + 1);
    return res;
}
};
struct Tag { // 有些 Tag 不用 push 例如 sweepLine
    void apply(const Tag &t) & {}
    Tag offset(int d) const { return *this; }
};
struct Info {
    void apply(const Tag &t, int l, int r) & {} /*
    Info &operator=(const Info &i) & {
        // do something... 部分 assignment 使用
        return *this;
    } */
};
Info operator+(const Info &a, const Info &b) {
    Info c;
    return c;
}

```

3.6 Persistent Segment Tree [d41d8c]

```

template<class Info> struct PST {
    struct Node {
        Info info = Info();
        int lc = 0, rc = 0;
    };
    int n;
    vector<Node> nd;
    vector<int> rt;
    PST(int n) : n(n), nd(1), rt(1) {}
    PST(const vector<Info> &v) : n(v.size()), nd(1) {
        auto build = [&](auto &&self, int l, int r) -> int {
            int p = generate();
            if (r - l == 1) {
                nd[p].info = v[l];
                return p;
            }
            int m = (l + r) / 2;
            int lc = self(self, l, m);
            int rc = self(self, m, r);
            nd[p].lc = lc;
            nd[p].rc = rc;
            pull(nd[p]);
            return p;
        };
        rt.push_back(build(build, 0, n));
    }
    void pull(Node &p) { p.info = nd[p.lc].info + nd[p.rc].info; }
    int copy(int p) { // copy 一個 node
        nd.push_back(nd[p]);
        return nd.size() - 1;
    }
    int generate() { // 創立新 node
        nd.emplace_back();
        return nd.size() - 1;
    }
    void modify(int x, const Info &i, int ver = 0) {
        if (int(rt.size()) <= ver) rt.resize(ver + 1);
        rt[ver] = modify(x, i, 0, n, rt[ver]);
    }
    int modify(int x, const Info &i, int l, int r, int p) {
        p = p ? copy(p) : generate();
        if (r - l == 1) {
            nd[p].info = i;
            return p;
        }
        int m = (l + r) / 2;
        if (x < m) {
            int y = modify(x, i, l, m, nd[p].lc);
            nd[p].lc = y;
        } else {
            int y = modify(x, i, m, r, nd[p].rc);
            nd[p].rc = y;
        }
        pull(nd[p]);
        return p;
    }
    Info query(int ql, int qr, int ver = 0) {
        return query(ql, qr, 0, n, rt[ver]);
    }
    Info query(int ql, int qr, int l, int r, int p) {
        if (l >= qr || r <= ql || p == 0) return Info();
    }
};

```



```

    if (ql <= l && r <= qr) return nd[p].info;
    int m = (l + r) / 2;
    return query(ql, qr, l, m, nd[p].lc) + query(ql, qr, m, r, nd[p].rc);
}
void createVersion(int ver) { rt.push_back(rt[ver]); }
};

```

3.7 Kth-element [d41d8c]

```

template<class T> struct StaticKth : PST<int> {
    int dct(T x) {
        return lower_bound(s.begin(), s.end(), x) - s.begin();
    }
    vector<T> a, s; // array, sorted
    StaticKth(const vector<T> &a, const vector<T> &s) : a(a), s(s), PST<int>(s.size()) {
        assert(is_sorted(s.begin(), s.end()));
        for (int i = 0; i < a.size(); i++) {
            createVersion(i);
            add(dct(a[i]), 1, i + 1);
        }
    }
    int work(int a, int b, int l, int r, int k) {
        if (r - l == 1) return l;
        int x = nd[nd[b].lc].info - nd[nd[a].lc].info;
        int m = (l + r) / 2;
        if (x >= k) return work(nd[a].lc, nd[b].lc, l, m, k);
        else return work(nd[a].rc, nd[b].rc, m, r, k - x);
    }
    int work(int l, int r, int k) { // [l, r), k > 0
        return s[work(rt[l], rt[r], 0, n, k)];
    }
}; // Dynamic -> Fenwick 包動態開點值域線段樹

```

3.8 Link Cut Tree [d41d8c]

```

// 有用到 pathApply 才需要 apply 有關的
const int Mod = 51061;
struct Tag {
    ll add = 0, mul = 1;
    void apply(const Tag &t) {
        mul = mul * t.mul % Mod;
        add = (add * t.mul % Mod + t.add) % Mod;
    }
};
struct Info {
    int siz = 0; // 實鏈的長度
    ll val = 0, sum = 0;
    void apply(const Tag &t) {
        val = (val * t.mul % Mod + t.add) % Mod;
        sum = (sum * t.mul % Mod + t.add * siz % Mod) % Mod;
    }
    void pull(const Info &l, const Info &r) {
        siz = 1 + l.siz + r.siz;
        sum = (l.sum + r.sum + val) % Mod;
    }
};
struct LinkCutTree { // 1-based
    vector<Info> info;
    vector<Tag> tag;
    vector<array<int, 2>> ch;
    vector<int> p, rev;
    LinkCutTree(int n) : info(n + 1), tag(n + 1), ch(n + 1, {0, 0}), p(n + 1), rev(n + 1) {}
    bool isrt(int x) {
        return ch[p[x]][0] != x && ch[p[x]][1] != x;
    }
    int pos(int x) { // x 是其 par 的左/右
        return ch[p[x]][1] == x;
    }
    void applyRev(int x) {
        if (!x) return;
        swap(ch[x][0], ch[x][1]);
        rev[x] ^= 1;
    }
    void apply(int x, const Tag &t) {
        if (!x) return;
        info[x].apply(t);
        tag[x].apply(t);
    }
    void push(int x) {
        if (rev[x]) {
            applyRev(ch[x][0]);
            applyRev(ch[x][1]);
            rev[x] = 0;
        }
        apply(ch[x][0], tag[x]);
        apply(ch[x][1], tag[x]);
        tag[x] = Tag();
    }
    void pull(int x) {
        if (!x) return;
        info[x].pull(info[ch[x][0]], info[ch[x][1]]);
    }
    void pushAll(int x) {
        if (!isrt(x)) pushAll(p[x]);
        push(x);
    }
    void rotate(int x) { // x 與其 par 交換位置
        int f = p[x], r = pos(x);
        ch[f][r] = ch[x][!r];
    }
}

```

```

    if (ch[x][!r]) p[ch[x][!r]] = f;
    p[x] = p[f];
    if (!isrt(f)) ch[p[f]][pos(f)] = x;
    ch[x][!r] = f, p[f] = x;
    pull(f), pull(x);
}
void splay(int x) { // x 旋轉到當前的根
    pushAll(x);
    for (int f = p[x]; f = p[x], !isrt(x); rotate(x))
        if (!isrt(f)) rotate(pos(x) == pos(f) ? f : x);
}
// access(x), access(y) 可以回傳 LCA
int access(int x) { // 根到 x 換成實鏈
    int c;
    for (c = 0; x; c = x, x = p[x]) {
        splay(x);
        ch[x][1] = c;
        pull(x);
    }
    return c;
}
void makeRoot(int x) { // x 變成所在樹的根
    access(x), splay(x), applyRev(x);
}
int findRoot(int x) {
    access(x), splay(x);
    while (ch[x][0]) x = ch[x][0];
    splay(x); return x;
}
void split(int rt, int x) { // 以 rt 為根, x 為子樹根
    makeRoot(x), access(rt), splay(rt);
}
void link(int rt, int x) { // 以 rt 為根, x 為子樹根
    makeRoot(rt);
    access(x), splay(x);
    p[rt] = x;
    pull(x);
}
void cut(int rt, int x) { // 以 rt 為根, x 為子樹根
    split(rt, x);
    ch[rt][0] = p[x] = 0;
    pull(rt);
}
bool connected(int x, int y) {
    return findRoot(x) == findRoot(y);
}
bool neighbor(int x, int y) {
    if (!connected(x, y)) return false;
    split(x, y);
    return info[x].siz == 2; // psiz
}
void modify(int x, const Info &i) {
    access(x), splay(x);
    info[x] = i, pull(x);
}
void pathApply(int x, int y, const Tag &t) {
    assert(connected(x, y));
    split(x, y), apply(x, t);
}
Info pathQuery(int x, int y) {
    assert(connected(x, y));
    split(x, y); return info[x];
}
}

```

3.9 LCT Extension [d41d8c]

```

struct Info {
    int siz = 0, vsiz
        = 0, psiz = 0; // 真實子樹大小, 虛子樹大小, 實鏈長度
    ll val = 0, sum = 0, vsum = 0, psum = 0;
    void apply(const Tag &t) {
        val = (val * t.mul % Mod + t.add) % Mod;
        sum = (sum - psum + Mod) % Mod;
        psum = (psum * t.mul % Mod + t.add * psiz % Mod) % Mod;
        sum = (sum + psum) % Mod;
    }
    void pull(const Info &l, const Info &r) {
        psiz = 1 + l.psiz + r.psiz;
        siz = 1 + l.siz + r.siz + vsiz;
        psum = (l.psum + r.psum + val) % Mod;
        sum = (l.sum + r.sum + val + vsum) % Mod;
    }
    void modify(const Info &i) { // 才不會蓋掉 vsum, vsiz
        val = i.val;
    }
    // 下面的 modify 也要改, 不能直接 =
    void operator+=(const Info &i) { // 將實邊轉虛邊時加上資訊
        vsiz += i.siz;
        vsum = (vsum + i.sum) % Mod;
    }
    void operator-=(const Info &i) { // 將虛邊轉實邊時扣除資訊
        vsiz -= i.siz;
        vsum = (vsum - i.sum % Mod + Mod) % Mod;
    }
    Info asSubtree
        () const { // 當前節點作為 Virtual Subtree 根時的狀態
        Info res;
        res.siz = vsiz + 1;
        res.sum = (vsum + val) % Mod;
    }
}

```

```

    return res;
}
};

struct LinkCutTree { // 加在註解後面
    int access(int x) { // ...splay(x);
        if (ch[x][1]) info[x] += info[ch[x][1]]; // 原右子樹轉虛邊
        if (c) info[x] -= info[c]; // 虛邊轉新右子樹
    }
    void link(int rt, int x) { // ...p[rt] = x;
        info[x] += info[rt]; // rt 成為 x 的虛子樹
    }
    Info subtreeQuery
        (int rt, int x) { // 以 rt 當 root, x 的 subtree
        assert(connected(rt, x));
        makeRoot(rt);
        access(x), splay(x);
        return info[x].asSubtree();
    }
    Info componentQuery(int x) {
        makeRoot(x);
        return info[x];
    }
};

```

3.10 Treap [d41d8c]

```

template<class Info, class Tag = bool()>
struct Treap { // 0 -> initial root
    vector<Info> info;
    // vector<Tag> tag;
    vector<int> siz, par, rev, pri;
    vector<array<int, 2>> ch;
    Treap(int n) : info(n + 1), siz
        (n + 1), par(n + 1), rev(n + 1), pri(n + 1), ch(n + 1) {
        // tag.resize(n + 1);
        for (int i = 1; i <= n; i++)
            siz[i] = 1, pri[i] = gen();
    }
    // void apply(int t, const Tag &v) {
    //     info[t].apply(siz[t], v);
    //     tag[t].apply(v);
    // }
    void push(int t) {
        if (rev[t]) {
            swap(ch[t][0], ch[t][1]);
            if (ch[t][0]) rev[ch[t][0]] ^= 1;
            if (ch[t][1]) rev[ch[t][1]] ^= 1;
            rev[t] = 0;
        }
        // apply(ch[t][0], tag[t]);
        // apply(ch[t][1], tag[t]);
        // tag[t] = Tag();
    }
    void pull(int t) {
        siz[t] = 1 + siz[ch[t][0]] + siz[ch[t][1]];
        info[t].pull(info[ch[t][0]], info[ch[t][1]]);
    }
    int merge(int a, int b) {
        if (!a || !b) return a ? a : b;
        push(a), push(b);
        if (pri[a] > pri[b]) {
            ch[a][1] = merge(ch[a][1], b);
            pull(a); return a;
        } else {
            ch[b][0] = merge(a, ch[b][0]);
            pull(b); return b;
        }
    }
    pair<int, int> split(int t, int k) {
        if (!t) return {0, 0};
        push(t);
        if (siz[ch[t][0]] >= k) {
            auto [a, b] = split(ch[t][0], k);
            ch[t][0] = b, pull(t);
            return {a, t};
        } else {
            auto [a, b] = split(ch[t][1], k - siz[ch[t][0]] - 1);
            ch[t][1] = a, pull(t);
            return {t, b};
        }
    }
    template<class F> // 尋找區間內，第一個符合條件的
    int findFirst(int t, F &&pred) {
        if (!t) return 0;
        push(t);
        if (!pred(info[t])) return 0;
        int idx = findFirst(ch[t][0], pred);
        if (!idx)
            idx = 1 + siz[ch[t][0]] + findFirst(ch[t][1], pred);
        return idx;
    }
    int getPos(int rt, int t) { // get t's index in array
        int res = siz[t] + 1;
        while (t != rt) {
            int p = par[t];
            if (ch[p][1] == t) res += siz[ch[p][0]] + 1;
            t = p;
        }
        return res;
    }
    void getArray(int t, vector<Info> &a) {

```

```

        if (!t) return;
        push(t);
        getArray(ch[t][0], a);
        a.push_back(info[t]);
        getArray(ch[t][1], a);
    }
};

struct Tag {
    int setVal; ll add;
    void apply(const Tag &t) {
        if (t.setVal) {
            setVal = t.setVal;
            add = t.add;
        } else {
            add += t.add;
        }
    }
};

struct Info {
    ll val, sum;
    void apply(int siz, const Tag &t) {
        if (t.setVal) {
            val = t.setVal;
            sum = 1LL * siz * t.setVal;
        }
        val += t.add;
        sum += 1LL * siz * t.add;
    }
    void pull(const Info &l, const Info &r) {
        sum = val + l.sum + r.sum;
    }
};

```

3.11 KDTree [d41d8c]

```

struct Info {
    static constexpr int DIM = 2;
    array<int, DIM> x, L, R;
    int v = 0, sum = 0;
    void pull(const Info &l, const Info &r) {
        sum = v + l.sum + r.sum;
    }
};

struct KDTree {
    static constexpr int DIM = Info::DIM;
    vector<Info> info;
    vector<int> rt, l, r, p;
    int n, lg;
    KDTree(int n) : n(n),
        lg(__lg(n)), info(1), rt(lg + 1), l(n + 1), r(n + 1) {}
    void pull(int p) {
        info[p].L = info[p].x;
        info[p].pull(info[l[p]], info[r[p]]);
        for (int ch : {l[p], r[p]}) {
            if (!ch) continue;
            for (int k = 0; k < DIM; k++) {
                info[p].L[k] = min(info[p].L[k], info[ch].L[k]);
                info[p].R[k] = max(info[p].R[k], info[ch].R[k]);
            }
        }
    }
    int rebuild(int l, int r, int dep = 0) {
        if (r == l) return 0;
        int m = (l + r) / 2;
        nth_element(p.begin() + l, p.begin() + m, p.begin() + r,
            [&](int x,
                int y) { return info[x].x[dep] < info[y].x[dep]; });
        int x = p[m];
        this->l[x] = rebuild(l, m, (dep + 1) % DIM);
        this->r[x] = rebuild(m + 1, r, (dep + 1) % DIM);
        pull(x);
        return x;
    }
    void append(int &x) {
        if (!x) return;
        p.push_back(x);
        append(l[x]);
        append(r[x]);
        x = 0;
    }
    void addNode(const Info &i) {
        p.assign(1, info.size());
        info.push_back(i);
        for (int j = 0; j++) {
            if (!rt[j]) {
                rt[j] = rebuild(0, p.size());
                break;
            } else {
                append(rt[j]);
            }
        }
    }
    Info query(int
        p, const array<int, DIM> &l, const array<int, DIM> &r) {
        if (!p) return Info();
        bool inside = true;
        for (int k = 0; k < DIM; k++) {
            inside &= (l[k] <= info[p].L[k] && info[p].R[k] <= r[k]);
        }
        if (inside) return info[p];
        for (int k = 0; k < DIM; k++) {
            if (info[p].R[k] < l[k] || r[k] < info[p].L[k]) {

```



```

        return Info();
    }
}
Info ans;
inside = true;
for (int k = 0; k < DIM; k++) {
    inside &= l[k] <= info[p].x[k] && info[p].x[k] <= r[k];
}
if (inside) ans = info[p];
ans.pull(query(this->l[p], l, r), query(this->r[p], l, r));
return ans;
}
Info query
(const array<int, DIM> &l, const array<int, DIM> &r) {
    Info res;
    for (int i = 0; i <= lg; i++) {
        res.pull(res, query(rt[i], l, r));
    }
    return res;
}
};

```

3.12 RMQ [d41d8c]

```

template<class T, class F = less<T>> struct RMQ { // [l, r)
    int n;
    F cmp = F();
    vector<vector<T>> g;
    RMQ() {}
    RMQ(const vector<T> &a, F cmp = F()) : cmp(cmp) {
        init(a);
    }
    void init(const vector<T> &a) {
        n = a.size();
        if (!n) return;
        int lg = __lg(n);
        g.resize(lg + 1);
        g[0] = a;
        for (int j = 1; j <= lg; j++) {
            g[j].resize(n - (1 << j) + 1);
            for (int i = 0; i <= n - (1 << j); i++)
                g[j][i] = min(g[j - 1][i], g[j - 1][i + (1 << (j - 1))], cmp);
        }
    }
    T operator()(int l, int r) {
        assert(0 <= l && l < r && r <= n);
        int lg = __lg(r - l);
        return min(g[lg][l], g[lg][r - (1 << lg)], cmp);
    }
};

```

3.13 Mo [d41d8c]

```

struct Query { int id, l, r; };
void mo(vector<Query> &q) {
    int blk = sqrt(q.size());
    sort(q
        .begin(), q.end(), [&](const Query &a, const Query &b) {
            int x = a.l / blk, y = b.l / blk;
            return x == y ? a.r < b.r : x < y;
        });
    int nl = 0, nr = -1;
    for (auto [id, l, r] : qry) {
        while (nr < r) nr++, addR();
        while (l < nl) nl--, addL();
        while (r < nr) delR(), nr--;
        while (nl < l) delL(), nl++;
    }
}

```

3.14 Dynamic Lazy Segment Tree [d41d8c]

```

template<class Info, class Tag> struct LazySegmentTreeDynamic {
    struct Node {
        Info info = Info();
        Tag tag = Tag();
        bool isLazy = false;
        int lc = 0, rc = 0;
    };
    int n, rt;
    vector<Node> nd;
    LazySegmentTreeDynamic(int n) : n(n), rt(0), nd(1) {}
    int generate() {
        nd.emplace_back();
        return nd.size() - 1;
    }
    void pull(int p) {
        nd[p].info = nd[nd[p].lc].info + nd[nd[p].rc].info;
    }
    void modify
        (int x, const Info &i) { rt = modify(x, i, 0, n, rt); }
    int modify(int x, const Info &i, int l, int r, int p) {
        if (p == 0) p = generate();
        if (r - l == 1) {
            nd[p].info = i;
            return p;
        }
        push(l, r, p);
        int m = (l + r) / 2;
        if (x < m) {
            int y = modify(x, i, l, m, nd[p].lc);
            nd[p].lc = y;
        }
    }
};

```

```

    } else {
        int y = modify(x, i, m, r, nd[p].rc);
        nd[p].rc = y;
    }
    pull(p);
    return p;
}
Info query
(int ql, int qr) { return query(ql, qr, 0, n, rt); }
Info query(int ql, int qr, int l, int r, int p) {
    if (l >= qr || r <= ql || p == 0) return Info();
    if (ql <= l && r <= qr) return nd[p].info;
    push(l, r, p);
    int m = (l + r) / 2;
    return query(ql, qr, l, m, nd[p].lc) + query(ql, qr, m, r, nd[p].rc);
}
void apply(const Tag &t, int l, int r, int p) {
    nd[p].info.apply(t, l, r);
    nd[p].tag.apply(t);
    nd[p].isLazy = true;
}
void push(int l, int r, int p) {
    if (!nd[p].isLazy) return;
    int m = (l + r) / 2;
    if (nd[p].lc == 0) nd[p].lc = generate();
    if (nd[p].rc == 0) nd[p].rc = generate();
    apply(nd[p].tag, l, m, nd[p].lc);
    apply(nd[p].tag.offset(m - l), m, r, nd[p].rc);
    nd[p].tag = Tag();
    nd[p].isLazy = false;
}
void rangeApply(int ql, int qr,
    const Tag &t) { rt = rangeApply(ql, qr, t, 0, n, rt); }
int rangeApply
(int ql, int qr, const Tag &t, int l, int r, int p) {
    if (l >= qr || r <= ql) return p;
    if (p == 0) p = generate();
    if (ql <= l && r <= qr) {
        apply(t.offset(l - ql), l, r, p);
        return p;
    }
    push(l, r, p);
    int m = (l + r) / 2;
    nd[p].lc = rangeApply(ql, qr, t, l, m, nd[p].lc);
    nd[p].rc = rangeApply(ql, qr, t, m, r, nd[p].rc);
    pull(p);
    return p;
}
template<class T> int findFirst(int ql, int
    qr, T &&f) { return findFirst(ql, qr, f, 0, n, rt); }
template<class T> int
    findFirst(int ql, int qr, T &&f, int l, int r, int p) {
        if (l >=
            qr || r <= ql || p == 0 || !f(nd[p].info)) return -1;
        if (r - l == 1) return l;
        int m = (l + r) / 2;
        push(l, r, p);
        int res = findFirst(ql, qr, f, l, m, nd[p].lc);
        if (res != -1) return res;
        return findFirst(ql, qr, f, m, r, nd[p].rc);
    }
};

```

4 Flow Matching

4.1 Dinic [d41d8c]

```

template<class T> struct Dinic {
    // augment time: O(VE), O(E) for unit capacity,
    // augment number: min(O(F), O(V))
    // min(O(E^0.5), O(V^2/3)) for unit capacity
    // O(V^0.5) for deg_in(u) or deg_out(u) <= 1
    // so bipartite matching: O(EV^0.5)
    struct Edge {
        int to;
        T f, cap; // 流量跟容量
    };
    int n, m, s, t;
    const T INF_FLOW = numeric_limits<T>::max() / 2;
    vector<vector<int>> g;
    vector<Edge> e;
    vector<int> h, cur;
    Dinic(int n) : n(n), m(0), g(n), h(n), cur(n) {}
    void addEdge(int u, int v, T cap) {
        e.push_back({v, 0, cap});
        e.push_back({u, 0, 0});
        g[u].push_back(m++);
        g[v].push_back(m++);
    }
    bool bfs() {
        fill(h.begin(), h.end(), -1);
        h[s] = 0; queue<int> q;
        q.push(s);
        while (!q.empty()) {
            int u = q.front(); q.pop();
            for (int id : g[u]) {
                auto [v, f, cap] = e[id];
                if (f != cap && h[v] == -1) {
                    h[v] = h[u] + 1;
                    if (v == t) return true;
                    q.push(v);
                }
            }
        }
    }
};

```

```

    }
}
return false;
}
T dfs(int u, T flow) {
    if (flow == 0) return 0;
    if (u == t) return flow;
    for (int &i = cur[u]; i < g[u].size(); i++) {
        int j = g[u][i];
        auto [v, f, cap] = e[j];
        if (f == cap || h[u] + 1 != h[v]) continue;
        T mn = dfs(v, min(flow, cap - f));
        if (mn > 0) {
            e[j].f += mn;
            e[j ^ 1].f -= mn;
            return mn;
        }
    }
    return T{};
}
T work(int s_, int t_) {
    s = s_; t = t_; T f = 0;
    while (bfs()) {
        fill(cur.begin(), cur.end(), 0);
        while (true) {
            T res = dfs(s, INF_FLOW);
            if (res == 0) break;
            f += res;
        }
    }
    return f;
}
void reuse(int n_) { // 走殘留網路, res += f
    while (n < n_) {
        g.emplace_back();
        h.emplace_back();
        cur.emplace_back();
        n += 1;
    }
}
};

```

4.2 Min Cut [d41d8c]

```

void policeChase(int n, Dinic<int> &d) {
    int ans = d.work(0, n - 1);
    vector<int> vis(n);
    [&](this auto &&self, int u) -> void {
        if (vis[u]) return;
        vis[u] = 1;
        for (int id : d.g[u]) {
            auto [to, f, cap] = d.e[id];
            if (cap - f > 0) self(to);
        }
    }(0);
    for (int i = 0; i < n; i++) {
        if (!vis[i]) continue;
        for (int id : d.g[i]) {
            if (id & 1) continue;
            auto e = d.e[id];
            if (!vis[e.to]) cout << i + 1 << " " << e.to + 1 << "\n";
        }
    }
}

```

4.3 MCMF [d41d8c]

```

template<class Tf, class Tc> struct MCMF {
    struct Edge {
        int to;
        Tf f, cap; // 流量跟容量
        Tc cost;
    };
    int n, m, s, t;
    const Tf INF_FLOW = numeric_limits<Tf>::max() / 2;
    const Tc INF_COST = numeric_limits<Tc>::max() / 2;
    vector<Edge> e;
    vector<vector<int>> g;
    vector<Tc> dis, pot;
    vector<int> rt, inq;
    MCMF(int n) : n(n), m(0), g(n) {}
    void addEdge(int u, int v, Tf cap, Tc cost) {
        e.push_back({v, 0, cap, cost});
        e.push_back({u, 0, 0, -cost});
        g[u].push_back(m++);
        g[v].push_back(m++);
    }
    bool spfa() { // O(FVE)
        dis.assign(n, INF_COST);
        rt.assign(n, -1), inq.assign(n, 0);
        queue<int> q; q.push(s);
        dis[s] = 0, inq[s] = 1;
        while (!q.empty()) {
            int u = q.front(); q.pop();
            inq[u] = 0;
            for (int id : g[u]) {
                auto [v, f, cap, cost] = e[id];
                Tc ndis = dis[u] + cost + pot[u] - pot[v];
                if (f < cap && dis[v] > ndis) {
                    dis[v] = ndis, rt[v] = id;
                    if (!inq[v])

```

```

                        q.push(v), inq[v] = 1;
                }
            }
            return dis[t] != INF_COST;
        }
    }
    bool dijkstra() { // O(FElogV)
        dis.assign(n, INF_COST), rt.assign(n, -1);
        priority_queue<pair<Tc, int>, vector<pair<Tc, int>>, greater<pair<Tc, int>>> pq;
        dis[s] = 0; pq.emplace(dis[s], s);
        while (!pq.empty()) {
            auto [d, u] = pq.top(); pq.pop();
            if (dis[u] < d) continue;
            for (int id : g[u]) {
                auto [v, f, cap, cost] = e[id];
                Tc ndis = dis[u] + cost + pot[u] - pot[v];
                if (f < cap && dis[v] > ndis) {
                    dis[v] = ndis, rt[v] = id;
                    pq.emplace(ndis, v);
                }
            }
        }
        return dis[t] != INF_COST;
    }
    pair<Tf, Tc> work(int s_, int t_, Tf need) {
        s = s_, t = t_; pot.assign(n, 0);
        Tf flow{}; Tc cost{}; int fr = 0;
        while (fr++ ? dijkstra() : spfa()) {
            for (int i = 0; i < n; i++)
                dis[i] += pot[i] - pot[s];
            Tf f = need;
            for (int i = t; i != s; i = e[rt[i] ^ 1].to)
                f = min(f, e[rt[i]].cap - e[rt[i]].f);
            for (int i = t; i != s; i = e[rt[i] ^ 1].to)
                e[rt[i]].f += f, e[rt[i] ^ 1].f -= f;
            flow += f, need -= f;
            cost += f * dis[t];
            swap(dis, pot);
            if (need == 0) break;
        }
        return {flow, cost};
    }
    void reset() {
        for (int i = 0; i < m; i++) e[i].f = 0;
    }
};

```

4.4 Hungarian [d41d8c]

```

struct Hungarian { // 0-based, O(VE)
    int n, m;
    vector<vector<int>> adj;
    vector<int> used, vis;
    vector<pair<int, int>> match;
    Hungarian(int n, int m) : n(n), m(m) {
        adj.assign(n + m, {});
        used.assign(n + m, -1);
        vis.assign(n + m, 0);
    }
    void addEdge(int u, int v) {
        adj[u].push_back(n + v);
        adj[n + v].push_back(u);
    }
    bool dfs(int u) {
        int sz = adj[u].size();
        for (int i = 0; i < sz; i++) {
            int v = adj[u][i];
            if (vis[v] == 0) {
                vis[v] = 1;
                if (used[v] == -1 || dfs(used[v])) {
                    used[v] = u;
                    return true;
                }
            }
        }
        return false;
    }
    vector<pair<int, int>> work() {
        match.clear();
        used.assign(n + m, -1);
        vis.assign(n + m, 0);
        for (int i = 0; i < n; i++) {
            fill(vis.begin(), vis.end(), 0);
            dfs(i);
        }
        for (int i = n; i < n + m; i++)
            if (used[i] != -1)
                match.emplace_back(used[i], i - n);
        return match;
    }
};

```

- 有向無環圖:
 - 最小不相交路徑覆蓋: 最小路徑數 = 頂點數 - 最大匹配數
 - 最小相交路徑覆蓋: 先用 Floyd 求傳遞封包, 有連邊就建邊, 然後再套最小不相交路徑覆蓋
- 二分圖:
 - 最小點覆蓋: 選出一些點, 使每條邊至少有一個端點被選到的最少數量
 - 最小點覆蓋 = 最大匹配數

- 還原解: flow 的作法是從源點開始 dfs, 只走 $\text{cap} - \text{flow} > 0$ 的邊, 最後挑選左邊還沒被跑過的點和右邊被跑過的點當作覆蓋的點
- 最大獨立集: 選出一些點, 使這些點兩兩沒有邊連接的最大數量
- 最大獨立集 = 點數 - 最大匹配數
- 還原解: 最小點覆蓋的補集
- 最少邊覆蓋: 選出一些邊, 讓所有點都覆蓋到的最少數量
- 最少邊覆蓋 = 點數 - 孤立點數 - 最大匹配數
- 還原解: 最大匹配 + 未被匹配的非孤立點隨便配

5 String

5.1 Hash [234076]

```
const int D = 59;
vector<int> rollingHash(string &s) {
    vector<int> a {0};
    for (auto c : s)
        a.push_back(mul(a.back(), D) + (c - 'A' + 1));
    return a;
}
int qryHash(vector<int> &h, int l, int r) { // [l, r)
    return sub(h[r], mul(h[l], power(D, r - l)));
}
```

5.2 KMP [cdf28b]

```
template<class T> struct KMP {
    int n;
    T pat;
    vector<int> f;
    // f 存匹配失敗時, 移去哪
    // 也就是 pat[0, i] 的最長共同前後綴長度
    // ex : a b c a b c
    //      -1 -1 -1 0 1 2
    KMP(const T &pat) : n(pat.size()), pat(pat), f(n, -1) {
        for (int i = 1; i < n; i++) {
            int j = f[i - 1];
            while (j != -1 && pat[j + 1] != pat[i]) j = f[j];
            if (pat[j + 1] == pat[i]) f[i] = j + 1;
        }
    }
    vector<int> work(const T &s) {
        vector<int> pos;
        for (int i = 0, j = -1; i < s.size(); i++) {
            while (j != -1 && pat[j + 1] != s[i]) j = f[j];
            if (pat[j + 1] == s[i]) j++;
            if (j + 1 == n) {
                pos.push_back(i - j);
                j = f[j];
            }
        }
        return pos;
    }
};
```

5.3 Z Function [a1f65d]

```
// z[i] 表示 s 和 s[i, n - 1] (以 s[i] 開頭的后綴)
// 的最長公共前綴 (LCP) 的長度
template<class T> vector<int> Z(const T &s) {
    int n = s.size();
    vector<int> z(n);
    z[0] = n; // lcp(s, s), -1 or n
    for (int i = 1, j = 1; i < n; i++) {
        z[i] = max(0, min(j + z[j] - i, z[i - j]));
        while (i + z[i] < n && s[z[i]] == s[i + z[i]]) z[i]++;
        if (i + z[i] > j + z[j]) j = i;
    }
    return z;
}
```

5.4 Manacher [9f76ee]

```
// 找到對於每個位置的迴文半徑
template<class T> vector<int> manacher(const T &s) {
    vector<int> t {-1};
    for (auto c : s) t.push_back(c), t.push_back(-1);
    int n = t.size();
    vector<int> r(n);
    for (int i = 0,
        j = 0; i < n; i++) { // i 是中心, j 是最長回文字串中心
        if (2 * j - i >= 0 && j + r[j] > i)
            r[i] = min(r[2 * j - i], j + r[j] - i);
        while (i - r[i]
            i >= 0 && i + r[i] < n && t[i - r[i]] == t[i + r[i]])
            r[i]++;
        if (i + r[i] > j + r[j]) j = i;
    }
    return r;
}
// # a # b # a #
// 1 2 1 4 1 2 1
// # a # b # b # a #
// 1 2 1 2 5 2 1 2 1
// 值 -1 代表原回文字串長度
// (id - val + 1) / 2 可得原字串回文開頭
```

5.5 Trie [2f0f82]

```
const int N = 1E7; // 0 -> initial state
const int ALPHABET_SIZE = 26;
int tot = 0;
int trie[N][ALPHABET_SIZE], cnt[N];
void reset() {
    tot = 0, fill_n(trie[0], ALPHABET_SIZE, 0);
}
int newNode() {
    int x = ++tot;
    cnt[x] = 0, fill_n(trie[x], ALPHABET_SIZE, 0);
    return x;
}
void add(const string &s, int i = 0, int p = 0) {
    if (i == s.size()) {
        cnt[p]++;
        return;
    }
    int &q = trie[p][s[i] - 'a'];
    if (!q) q = newNode();
    add(s, i + 1, q);
}
int find(const string &s, int i = 0, int p = 0) {
    if (i == s.size()) return cnt[p];
    int q = trie[p][s[i] - 'a'];
    if (!q) return 0;
    return find(s, i + 1, q);
}
```

5.6 SA [9262af]

```
template<class T> struct SuffixArray {
    int n;
    vector<int> sa, rk, lc;
    // n: 字串長度
    // sa: 後綴數組, sa[i] 表示第 i 小的後綴的起始位置
    // rk: 排名數組, rk[i] 表示從位置 i 開始的後綴的排名
    // lc: LCP
    // 數組, lc[i] 表示 sa[i] 和 sa[i + 1] 的最長公共前綴長度
    SuffixArray(
        (const T &s) : n(s.size()), sa(n), lc(n - 1), rk(n) {
        iota(sa.begin(), sa.end(), 0);
        sort(sa.begin(),
            sa.end(), [&](int a, int b) { return s[a] < s[b]; });
        rk[sa[0]] = 0;
        for (int i = 1; i < n; i++)
            rk[sa[i]] = rk[sa[i - 1]] + (s[sa[i]] != s[sa[i - 1]]);
        int k = 1;
        vector<int> tmp, cnt(n);
        tmp.reserve(n);
        while (rk[sa[n - 1]] < n - 1) {
            tmp.clear();
            for (int i = 0; i < k; i++) tmp.push_back(n - k + i);
            for (auto i : sa) if (i >= k) tmp.push_back(i - k);
            fill(cnt.begin(), cnt.end(), 0);
            for (int i = 0; i < n; i++) cnt[rk[i]]++;
            for (int i = 1; i < n; i++) cnt[i] += cnt[i - 1];
            for (int i =
                n - 1; i >= 0; i--) sa[--cnt[rk[tmp[i]]]] = tmp[i];
            swap(rk, tmp); rk[sa[0]] = 0;
            for (int i = 1; i < n; i++) rk[sa[i]] = rk[
                sa[i - 1]] + (tmp[sa[i - 1]] < tmp[sa[i]] || sa[i -
                1] + k == n || tmp[sa[i - 1]] + k < tmp[sa[i] + k]);
            k *= 2;
        }
        for (int i = 0, j = 0; i < n; i++) {
            if (rk[i] == 0) {
                j = 0;
            } else {
                for (j -= j > 0; i + j < n && sa[rk[i] - 1]
                    + j < n && s[i + j] == s[sa[rk[i] - 1] + j]; j++);
                lc[rk[i] - 1] = j;
            }
        }
    }
};
RMQ<int> rmq(sa.lc);
auto lcp = [&](int i, int j) { // [i, j]
    i = sa.rk[i], j = sa.rk[j];
    if (i > j) swap(i, j);
    assert(i != j);
    return rmq(i, j);
};
```

5.7 AC [d34486]

```
struct AC {
    static constexpr int ALPHABET_SIZE = 26;
    struct Node {
        int fail; // 指向最長後綴
        int cnt; // 有多少模式字串是自己的後綴
        array<int, ALPHABET_SIZE> ch, next;
        // next 是補全後的轉移
    };
    vector<Node> t;
    AC() : t(1) {}
    int newNode() {
        t.emplace_back();
        return t.size() - 1;
    }
}
```

```

int insert(const string &s) {
    int u = 0;
    for (char c : s) {
        if (!t[u].ch[c - 'a']) {
            int v = newNode();
            t[u].ch[c - 'a'] = v;
        }
        u = t[u].ch[c - 'a'];
    }
    t[u].cnt++;
    return u;
}

void build() {
    queue<int> q;
    for (int c = 0; c < ALPHABET_SIZE; c++) {
        if (t[0].ch[c]) {
            q.push(t[0].ch[c]);
            t[0].next[c] = t[0].ch[c];
        }
    }
    while (!q.empty()) {
        int u = q.front(); q.pop();
        for (int c = 0; c < ALPHABET_SIZE; c++) {
            if (t[u].ch[c]) {
                int v = t[u].ch[c];
                t[v].fail = t[t[u].fail].next[c];
                t[v].cnt += t[t[u].fail].cnt;
                t[u].next[c] = v;
            } else {
                t[u].next[c] = t[t[u].fail].next[c];
            }
        }
    }
}
};

```

5.8 SAM [83a519]

```

struct SAM {
    // 0 -> initial state
    static constexpr int ALPHABET_SIZE = 26;
    // node -> strings with the same endpos set
    // link -> longest suffix with different endpos set
    // len -> state's longest suffix
    // fpos -> first endpos
    // strlen range -> [len(link) + 1, len]
    struct Node {
        int len, link = -1, fpos;
        array<int, ALPHABET_SIZE> next;
    };
    vector<Node> t;
    SAM() : t(1) {}
    int newNode() {
        t.emplace_back();
        return t.size() - 1;
    }
    int extend(int p, int c) {
        int cur = newNode();
        t[cur].len = t[p].len + 1;
        t[cur].fpos = t[cur].len - 1;
        while (p != -1 && !t[p].next[c]) {
            t[p].next[c] = cur;
            p = t[p].link;
        }
        if (p == -1) {
            t[cur].link = 0;
        } else {
            int q = t[p].next[c];
            if (t[p].len + 1 == t[q].len) {
                t[cur].link = q;
            } else {
                int r = newNode();
                t[r] = t[q];
                t[r].len = t[p].len + 1;
                while (p != -1 && t[p].next[c] == q) {
                    t[p].next[c] = r;
                    p = t[p].link;
                }
                t[q].link = t[cur].link = r;
            }
        }
        return cur;
    }
};

void solve(int n, string s, ll k) { // Substring Order II
    vector<int> last(n + 1);
    SAM sam;
    for (int i = 0; i < n; i++)
        last[i + 1] = sam.extend(last[i], s[i] - 'a');
    int sz = sam.t.size();

    vector<int> cnt(sz); // endpos size
    for (int i = 1; i <= n; i++) cnt[last[i]]++;
    vector<vector<int>> g(sz);
    for (int i = 1; i < sz; i++)
        g[sam.t[i].link].push_back(i);
    [&](this auto &&self, int u) -> void {
        for (auto v : g[u]) self(v), cnt[u] += cnt[v];
    } (0);

    vector<ll> dp(sz, -1);

```

```

// for any path from
// root, how many substring's prefix is the the path string
[&](this auto &&self, int u) -> ll {
    if (dp[u] != -1) return dp[u];
    dp[u] = cnt[u]; // distinct: = 1
    for (int c = 0; c < SAM::ALPHABET_SIZE; c++) {
        int v = sam.t[u].next[c];
        if (v) dp[u] += self(v);
    }
    return dp[u];
} (0);

int p = 0; string ans;
while (k > 0) { // 1-based
    for (int c = 0; c < SAM::ALPHABET_SIZE; c++) {
        int v = sam.t[p].next[c];
        if (v) {
            if (k > dp[v]) {
                k -= dp[v];
            } else {
                ans.push_back('a' + c);
                k -= cnt[v]; // distinct: --
                p = v; break;
            }
        }
    }
} cout << ans << "\n";
}

```

5.9 Palindrome Tree [e5a1ed]

```

struct PAM {
    // 0 -> even root, 1 -> odd root
    static constexpr int ALPHABET_SIZE = 26;
    // fail -> longest prefix(suffix) palindrome
    // number end at i = end at link[last[i]] + 1
    struct Node {
        int len, fail, cnt;
        array<int, ALPHABET_SIZE> next;
        Node() : len{}, fail{}, next{} {}
    };
    vector<int> s;
    vector<Node> t;
    PAM() {
        t.assign(2, Node());
        t[0].len = 0, t[0].fail = 1;
        t[1].len = -1;
    }
    int newNode() {
        t.emplace_back();
        return t.size() - 1;
    }
    int getFail(int p, int i) {
        while (i - t[p].len < 1 || s[i - t[p].len - 1] != s[i])
            p = t[p].fail;
        return p;
    }
    int extend(int p, int c) {
        int i = s.size();
        s.push_back(c);
        p = getFail(p, i);
        if (!t[p].next[c]) {
            int r = newNode();
            int v = getFail(t[p].fail, i);
            t[r].len = t[p].len + 2;
            t[r].fail = t[v].next[c];
            t[p].next[c] = r;
        }
        return p = t[p].next[c];
    }
};

void solve() {
    string s; cin >> s;
    int n = s.length();
    vector<int> last(n + 1);
    last[0] = 1;
    PAM pam;
    for (int i = 0; i < n; i++)
        last[i + 1] = pam.extend(last[i], s[i] - 'a');
    int sz = pam.t.size();
    vector<int> cnt(sz);
    for (int i = 1; i <= n; i++)
        cnt[last[i]]++; // 去重 = 1
    for (int i = sz - 1; i > 1; i--)
        cnt[pam.t[i].fail] += cnt[i];
}

```

5.10 Duval [f0a5c2]

```

// duval_algorithm
// 將字串分解成若干個非嚴格遞減的非嚴格遞增字串
template<class T> vector<T> duval(const T &s) {
    int i = 0, n = s.size();
    vector<T> res;
    while (i < n) {
        int k = i, j = i + 1;
        for (; j < n && s[k] <= s[j]; j++) {
            if (s[k] < s[j]) k = i;
            else k++;
        }
        while (i <= k) {
            res.emplace_back(s.begin() + i, s.begin() + i + j - k);

```

```

        i += j - k;
    }
}
return res;
}
// 最小旋轉字串
template<class T> T minRound(T s) {
    s.insert(s.end(), s.begin(), s.end());
    int i = 0, n = s.size(), start = i;
    while (i < n / 2) {
        start = i;
        int k = i, j = i + 1;
        for (; j < n && s[k] <= s[j]; j++) {
            if (s[k] < s[j]) k = i;
            else k++;
        }
        while (i <= k) i += j - k;
    }
    return {s.begin() + start, s.begin() + start + n / 2};
}

```

6 Math

6.1 Mint [b6abd5]

```

const int P = 1E9 + 7;
ll mul(ll a, ll b, ll p) { // P 超過 int 再用，慢
    ll res = a * b - ll(1.L * a * b / p) * p;
    res %= p;
    if (res < 0) res += p;
    return res;
}
template<class T> constexpr T power(T a, ll b) {
    T res{1};
    for (; b > 0; b >= 1, a = a * a)
        if (b & 1) res = res * a;
    return res;
}
template<int P> struct Mint {
    static int Mod;
    static int p() { return P > 0 ? P : Mod; }
    static void setMod(int Mod_) { Mod = Mod_; }
    int x;
    Mint(ll v = 0) : x(v % p()) { if (x < 0) x += p(); }
    explicit operator int() const { return x; }
    Mint operator-() const { return p() - x; }
    Mint inv() const { return power(*this, p() - 2); }

    Mint &operator+=(Mint a) { if ((x += a.x) >= p()) x -= p(); return *this; }
    Mint &operator-=(Mint a) { if ((x -= a.x) < 0) x += p(); return *this; }
    Mint &operator*=(Mint a) { x = 1LL * x * a.x % p(); return *this; }
    Mint &operator/=(Mint a) { return *this *= a.inv(); }

    friend Mint operator+(Mint a, Mint b) { return a + b; }
    friend Mint operator-(Mint a, Mint b) { return a - b; }
    friend Mint operator*(Mint a, Mint b) { return a * b; }
    friend Mint operator/(Mint a, Mint b) { return a / b; }

    bool operator==(Mint y) const { return x == y.x; }
    bool operator!=(Mint y) const { return x != y.x; }

    friend istream &operator>>(istream &is, Mint &a)
    { ll v; is >> v; a = Mint(v); return is; }
    friend ostream &operator<<(ostream &os, Mint a)
    { return os << a.x; }
};
template<> int Mint<0>::Mod = 998244353;
using Z = Mint<P>;

vector<Z> fac, invfac, inv;
void init(int n) {
    fac.resize(n + 1);
    invfac.resize(n + 1);
    inv.resize(n + 1);
    fac[0] = invfac[0] = 1;
    for (int i = 1; i <= n; i++) {
        fac[i] = fac[i - 1] * i;
    }
    invfac[n] = fac[n].inv();
    for (int i = n; i > 0; i--) {
        invfac[i - 1] = invfac[i] * i;
        inv[i] = invfac[i] * fac[i - 1];
    }
}
Z binom(int n, int m) {
    if (n < m || m < 0) return 0;
    return fac[n] * invfac[m] * invfac[n - m];
}
Z lucas(ll n, ll m) { // O(p + T log(n)), p is prime
    return m ? binom(n % P, m % P) * lucas(n / P, m / P) : 1;
}

```

6.2 Sieve [ad6c6e]

```

vector<int> minp, primes, mu, phi, dnum, pnum;
vector<ll> dsum;
void sieve(int n) {
    minp.assign(n + 1, 0);

```

```

    phi.assign(n + 1, 0), mu.assign(n + 1, 0);
    phi[1] = mu[1] = 1;
    for (int i = 2; i <= n; i++) {
        if (!minp[i]) {
            minp[i] = i;
            primes.push_back(i);
            phi[i] = i - 1;
            mu[i] = -1;
        }
        for (int p : primes) {
            if (i * p > n) break;
            minp[i * p] = p;
            if (p == minp[i]) {
                phi[i * p] = phi[i] * p;
                mu[i * p] = 0;
                break;
            } else {
                phi[i * p] = phi[i] * (p - 1);
                mu[i * p] = -mu[i];
            }
        }
    }
    dnum.assign(n + 1, 0), dsum.assign(n + 1, 0);
    for (int i = 1; i <= n; i++)
        for (int j = i; j <= n; j += i) dnum[j]++, dsum[j] += i;
    pnum.assign(n + 1, 0);
    for (int i : primes)
        for (int j = i; j <= n; j += i) pnum[j]++;
}

```

6.3 Matrix [6b2cbc]

```

using Matrix = vector<vector<Z>>;
Matrix operator*(const Matrix &a, const Matrix &b) {
    int n = a.size(), k = a[0].size(), m = b[0].size();
    assert(k == b.size());
    Matrix res(n, vector<Z>(m));
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++)
            for (int l = 0; l < k; l++)
                res[i][j] += a[i][l] * b[l][j];
    return res;
}
Matrix power(Matrix a, ll b) {
    int n = a.size();
    Matrix res(n, vector<Z>(n));
    for (int i = 0; i < n; i++) res[i][i] = 1;
    for (; b > 0; b >= 1, a = a * a)
        if (b & 1) res = res * a;
    return res;
}

```

6.4 Miller Rabin Pollard Rho [394cfb]

```

ll mul(ll a, ll b, ll p) {
    ll res = a * b - ll(1.L * a * b / p) * p;
    res %= p;
    if (res < 0) res += p;
    return res;
}
ll power(ll a, ll b, ll p) {
    ll res{1};
    for (; b > 0; b /= 2, a = mul(a, a, p))
        if (b & 1) res = mul(res, a, p);
    return res;
}
vector<ll>
> chk {2, 325, 9375, 28178, 450775, 9780504, 1795265022};
bool check(ll a, ll d, int s, ll n) {
    a = power(a, d, n);
    if (a == 1) return 1;
    for (int i = 0; i < s; i++, a = mul(a, a, n)) {
        if (a == 1) return 0;
        if (a == n - 1) return 1;
    }
    return 0;
}
bool isPrime(ll n) {
    if (n < 2) return 0;
    if (n % 2 == 0) return n == 2;
    ll d = n - 1, s = 0;
    while (d % 2 == 0) d /= 2, s++;
    for (ll i : chk)
        if (!check(i, d, s, n)) return 0;
    return 1;
}
const vector<ll> small = {2, 3, 5, 7, 11, 13, 17, 19};
ll findFactor(ll n) {
    if (isPrime(n)) return 1;
    for (ll p : small)
        if (n % p == 0) return p;
    ll x, y = 2, d, t = 1;
    auto f = [&](ll a) {
        return (mul(a, a, n) + t) % n;
    };
    for (int l = 2; ; l *= 2) {
        x = y;
        int m = min(l, 32);
        for (int i = 0; i < l; i += m) {
            d = 1;
            for (int j = 0; j < m; j++)
                y = f(y), d = mul(d, abs(x - y), n);

```

```

    ll g = __gcd(d, n);
    if (g == n) {
        l = 1, y = 2, ++t;
        break;
    }
    if (g != 1) return g;
}
}
map<ll, int> res;
void pollardRho(ll n) {
    if (n == 1) return;
    if (isPrime(n)) {
        res[n]++;
        return;
    }
    ll d = findFactor(n);
    pollardRho(n / d), pollardRho(d);
}
}

```

6.5 Primitive Root [b0ba96]

```

int findPrimitiveRoot(ll m) {
    Mlong<0>::setMod(m); // Mlong if needed
    ll phi = m; // m - 1 if m prime
    res.clear();
    pollardRho(m);
    for (auto [p, _] : res) phi = phi / p * (p - 1);
    vector<ll> pr; // prime factors of phi
    res.clear();
    pollardRho(phi);
    for (auto &[p, _] : res) pr.push_back(p);
    for (int i = 1; i <= m; i++) {
        bool ok = true;
        for (int j = 0; j < int(pr.size()) && ok; j++)
            ok &= power(Mlong<0>(i), phi / pr[j]).x != 1;
        if (ok) return i;
    }
    return -1; // m != 1, 2, 4, p^k, 2(p^k)
}

```

6.6 CRT [35109b]

```

// ax = b (mod m) 的前提是 gcd(a, m) | b
// Ax + By = C (g = gcd(A, B))
// (A/g) * x + (B/g) * y = C/g
// solve (A/g) * x' + (B/g) * y' = 1
// x = (x' * (C/g)) + i(B/g)
pair<ll, ll> exgcd(ll a, ll b) {
    if (b == 0) return {1, 0};
    auto [y, x] = exgcd(b, a % b);
    return {x, y - (a / b) * x};
} // 結果可能為負，注意正規化
// Smallest non-negative solution
using i128 = __int128_t;
pair<ll, ll> CRT(ll r1, ll m1, ll r2, ll m2) {
    ll g = __gcd(m1, m2);
    if ((r2 - r1) % g) return {-1, g};
    m1 /= g, m2 /= g;
    auto [p1, p2] = exgcd(m1, m2);
    i128 lcm = i128(m1) * m2 * g;
    i128 res = i128(p1) * (r2 - r1) * m1 + r1;
    return {(res % lcm + lcm) % lcm, lcm};
}
pair<ll, ll> EXCRT(vector<pair<int, int>> a) {
    ll R = 0, M = 1;
    for (auto [r, m] : a) {
        auto [res, lcm] = CRT(R, M, r, m);
        if (res == -1) return {-1, -1};
        R = res, M = lcm;
    }
    return {R, M};
}
// gcd(mod) = 1, support 3 1E9 Mod
pair<i128, i128> CRT(vector<pair<int, int>> a) {
    i128 s = 1, res = 0;
    for (auto [r, m] : a) s *= m;
    for (auto [r, m] : a) {
        i128 t = s / m;
        res = (res + r * t % s * exgcd(t, m).first % s) % s;
    }
    return {(res + s) % s, s};
}

```

6.7 XOR Basis [02f0c0]

```

auto add = [&](vector<int> &bas, int x) {
    for (auto i : bas) x = min(x, x ^ i);
    if (x) bas.push_back(x);
};
sort(bas.begin(), bas.end()); // 最簡化列梯
for (auto i = bas.begin(); i != bas.end(); i++) {
    for (auto j = next(i); j != bas.end(); j++) {
        *j = min(*j, *j ^ *i);
    }
}
// [l, r] 的區間 xor max (1-indexed)
vector<array<int, B>> bas(n + 1), lst(n + 1);
for (int i = 1; i <= n; i++) {
    int x = a[i - 1], p = i;
    bas[i] = bas[i - 1], lst[i] = lst[i - 1];
}

```

```

for (int j = B - 1; j >= 0; j--)
    if (x >> j & 1) {
        if (!bas[i][j]) {
            bas[i][j] = x, lst[i][j] = p;
            break;
        }
        if (lst[i][j] < p)
            swap(lst[i][j], p), swap(x, bas[i][j]);
        x ^= bas[i][j];
    }
}
auto qry = [&](int l, int r) -> int {
    int mx = 0;
    for (int i = B - 1; i >= 0; i--)
        if (lst[r][i] >= l)
            mx = max(mx, mx ^ bas[r][i]);
    return mx;
};

```

6.8 Gaussian Elimination [5d1aa7]

```

// 找反矩陣
    就開 2n，右邊放單位矩陣，做完檢查左半是不是單位，回傳右半
// 0 : no solution
// -1 : infinity solution
// 1 : one solution
template<class T> tuple<T,
    int, vector<T>> gaussianElimination(vector<vector<T>> a) {
    T det = 1;
    bool zeroDet = false;
    int n = a.size(), m = a[0].size(), rk = 0, sgn = 1;
    for (int c = 0; c < n; c++) {
        int p = -1;
        for (int r = rk; r < n; r++) {
            if (a[r][c] != 0) {
                p = r;
                break;
            }
        }
        if (p == -1) {
            zeroDet = true;
            continue;
        }
        if (p != rk) swap(a[rk], a[p]), sgn *= -1;
        det *= a[rk][c];
        T inv = 1 / a[rk][c];
        for (int j = c; j < m; j++) a[rk][j] *= inv;
        for (int r = 0; r < n; r++) {
            if (r == rk || a[r][c] == 0) continue;
            T fac = a[r][c];
            for (int j = c; j < m; j++)
                a[r][j] -= fac * a[rk][j];
        }
        rk++;
    }
    det = (zeroDet ? 0 : det * sgn);
    for (int r = rk; r < n; r++)
        if (a[r][m - 1] != 0) return {det, 0, {}};
    if (rk < n) return {det, -1, {}};
    vector<T> ans(n);
    for (int i = 0; i < n; i++) ans[i] = a[i][m - 1];
    return {det, 1, ans};
}
template<class T> tuple<int, vector<T>, vector<vector<T>>> findBasis(vector<vector<T>> a) {
    int n = a.size(), m = a[0].size(), rk = 0;
    vector<int> pos(m - 1, -1);
    for (int c = 0; c < m - 1; c++) {
        int p = -1;
        for (int r = rk; r < n; r++) {
            if (a[r][c] != 0) {
                p = r;
                break;
            }
        }
        if (p == -1) continue;
        if (p != rk) swap(a[rk], a[p]);
        pos[c] = rk;
        T inv = 1 / a[rk][c];
        for (int j = c; j < m; j++) a[rk][j] *= inv;
        for (int r = 0; r < n; r++) {
            if (r == rk || a[r][c] == 0) continue;
            T fac = a[r][c];
            for (int j = c; j < m; j++)
                a[r][j] -= fac * a[rk][j];
        }
        rk++;
    }
    vector<T> sol(m - 1);
    vector<vector<T>> basis;
    for (int r = rk; r < n; r++)
        if (a[r][m - 1] != 0)
            return {-1, sol, basis};
    for (int c = 0; c < m - 1; c++)
        if (pos[c] != -1)
            sol[c] = a[pos[c]][m - 1];
    for (int c = 0; c < m - 1; c++)
        if (pos[c] == -1) {
            vector<T> v(m - 1);
            v[c] = 1;
            for (int j = 0; j < m - 1; j++)
                if (pos[j] != -1)

```



```

    v[j] = -a[pos[j]][c];
    basis.push_back(v);
}
return {rk, sol, basis};
}
template<class T> using Matrix = vector<vector<T>>>;

```

6.9 Integer Partition [83bc9d]

```

// CSES_Sum_of_Divisors
const int Mod = 1E9 + 7;
const int inv_2 = 500000004;
// n / 1 * 1 + n / 2 * 2 + n / 3 * 3 + ... + n / n * n
void integerPartition() {
    ll ans = 0, n; cin >> n;
    for (ll l = 1, r; l <= n; l = r + 1) {
        r = n / (n / l);
        ll val = n / l; // n / l 到 n / r 一樣的值
        ll sum = ((l + r) %
            Mod) * ((r - l + 1) % Mod) % Mod * inv_2; // l 加到 r
        val %= Mod; sum %= Mod;
        ans += val * sum;
        ans %= Mod;
    }
    cout << ans << "\n";
}

```

6.10 Basic Theorems

- $a^{m-1} \equiv 1 \pmod{m}$
- $a^{m-2} \equiv a^{-1} \pmod{m}$
- $\tau(N) = \prod_{i=1}^k (p_i^0 + p_i^1 + \dots + p_i^{a_i})$
- $\prod_{d|N} d = N^{\tau(N)}$
- Highly Composite Number:
 - $(10^6): 720720 = 2^4 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13, \tau = 240$
 - $(10^9): 735134400 = 2^6 \cdot 3^3 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17, \tau = 1344$
 - $(10^{12}): 963761198400 = 2^6 \cdot 3^4 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23, \tau = 6720$
 - $(10^{15}): 866421317361600 = 2^6 \cdot 3^4 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31, \tau = 26880$
 - $(10^{18}): 897612484786617600 = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7^2 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 37, \tau = 103680$

6.10.1 Combination

- $C(n, m) = C(n, m-1) * (n-m+1) / m$
- $C(n+1, m) = C(n, m) + C(n, m-1)$
- $C(n, k) \equiv 1 \pmod{2} \Leftrightarrow$ all bit of $k \leq$ all bit of n in binary
- $(x+y)^n = \sum_{k=0}^n C(n, k) x^{n-k} y^k$

6.10.2 Game Theorem

- sg 值為 0 代表先手必敗
- 當前 sg 值 = 經過一方操作後，另一方所有可能的後繼狀態的 mex
- 若某個後繼狀態由多個狀態構成，就 xor 起來當作此後繼狀態的 sg 值
- 單組基礎 nim 的 sg 值為本身的原因: $f(0) = 0, f(1) = \text{mex}(f(0)) = 1, f(2) = \text{mex}(f(0), f(1)) = 2, \dots$ 都是自己
- 多組賽局可以把 sg 值 xor 起來，當成最後的 sg 值，nim 也是一樣，且由於 xor 性質，如果可以快速知道 $sg(1)g(2)\dots g(n)$ ，就可以用 xor 性質處理不連續組合

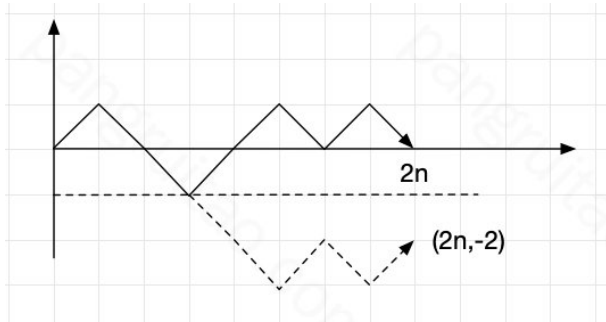
6.10.3 Pisano Period

- $\pi(ab) = \text{lcm}(\pi(a), \pi(b)) \quad (\gcd(a, b) = 1)$
- $\pi(p^e) | \pi(p) \cdot p^{e-1}$
- $\pi(p) | p^2 - 1 \quad (p \neq 2, 5)$
- $\pi(2) = 3, \pi(5) = 20$
- so can deal with $p \approx 10^9$ in long long

6.10.4 Burnside's lemma

- $|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$
- G : 各種翻轉操作所構成的置換群
- X/G : 本質不同的方案的集合
- X^g : 對於某一種操作 g ，所有方案中，經過 g 這種翻轉後保持不變的方案集合
- 集合取絕對值代表集合數

6.10.5 Catalan Theorem



- n 個往上 n 個往下，先枚舉所有情況 $\frac{(2n)!}{n!n!} = C_n^{2n}$
 - 扣掉非法的，有多少種可能讓最後的點落在 $(2n, -2)$
- 假設往上有 x 個，往下有 y 個，會有：

$$\begin{cases} x+y=2n \\ y-x=2 \end{cases} \Rightarrow \begin{cases} x=n-1 \\ y=n+1 \end{cases}$$

所以只要扣掉 C_{n-1}^{2n-1} 即可

6.10.6 Inversion

- 數論分塊可以快速計算一些含有除法向下取整的和式，就是像 $\sum_{i=1}^n f(i)g(\lfloor \frac{n}{i} \rfloor)$ 的和式。當可以在 $O(1)$ 內計算 $f(r) - f(l)$ 或已經預處理出 f 的前綴和時，數論分塊就可以在 $O(\sqrt{n})$ 的時間內計算上述和式的值。
- 迪利克雷捲積 $h(x) = \sum_{d|x} f(d)g(\frac{x}{d})$
- 積性函數
 - $\epsilon(n)$: 只在 $n=1$ 時為 1，其餘情況皆為 0。
 - 莫比烏斯函數

1.

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & \text{for } n=1 \\ 0 & \text{for } n \neq 1 \end{cases}$$

2. μ 是常數函數 1 的反元素

$$\Rightarrow \mu * 1 = \epsilon$$

- ϕ 歐拉函數: x 以下與 x 互質的數量

$$\phi(p_1^{a_1} \dots p_k^{a_k}) = (p_1^{a_1-1}(p_1-1)) \dots (p_k^{a_k-1}(p_k-1))$$

$$\star n = \sum_{d|n} \phi(d)$$

$\star$

$$\phi * 1 = \sum_{d|n} \phi(\frac{n}{d}) \quad \text{質因數分解}$$

$$= \sum_{i=0}^c \phi(p^i)$$

$$= 1 + p^0(p-1) + p^1(p-1) + \dots + p^{c-1}(p-1)$$

$$= p^c$$

$$= id$$

莫比烏斯反演公式

$$f(n) = \sum_{d|n} g(d) \Leftrightarrow g(n) = \sum_{d|n} \mu(d) f(\frac{n}{d})$$

$$f(n) = \sum_{n|d} g(d) \Leftrightarrow g(n) = \sum_{n|d} \mu(\frac{d}{n}) f(d)$$

例子

$$\sum_{i=a}^b \sum_{j=c}^d [gcd(i, j) = k]$$

$$\Rightarrow \sum_{i=1}^x \sum_{j=1}^y [gcd(i, j) = k]$$

$$= \sum_{i=1}^{\lfloor \frac{x}{k} \rfloor} \sum_{j=1}^{\lfloor \frac{y}{k} \rfloor} \epsilon(gcd(i, j))$$

$$= \sum_{i=1}^{\lfloor \frac{x}{k} \rfloor} \sum_{j=1}^{\lfloor \frac{y}{k} \rfloor} \sum_{d|gcd(i, j)} \mu(d)$$

$$= \sum_{d=1}^{\infty} \mu(d) \sum_{i=1}^{\lfloor \frac{x}{kd} \rfloor} \sum_{j=1}^{\lfloor \frac{y}{kd} \rfloor} [d|j] \quad d \text{ 可整除 } i \text{ 時為 } 1$$

$$= \sum_{d=1}^{\min(\lfloor \frac{x}{k} \rfloor, \lfloor \frac{y}{k} \rfloor)} \mu(d) \lfloor \frac{x}{kd} \rfloor \lfloor \frac{y}{kd} \rfloor$$

6.11 Mobius Inverse [d41d8c]

```

void solve() { // pref: pref of mu
    ll a, b, c, d, k; cin >> a >> b >> c >> d >> k;
    sieve(N);
    auto cal = [&](ll x, ll y) -> int {
        int res = 0;
        for (int l = 1, r; l <= min(x, y); l = r + 1) {
            r = min(x / (x / l), y / (y / l));
            res += (pref[r] - pref[l - 1]) * (x / l) * (y / l);
        }
        return res;
    };
    cout <<
        cal(b / k, d / k) - cal((a - 1) / k, d / k) - cal(b / k, (c - 1) / k) + cal((a - 1) / k, (c - 1) / k) << "\n";
}

```

6.12 LCM Convolution [a64dbb]

```

// ak = sum_{d|k} a[d] (zeta, a \ast 1)
// ck = sum_{d|k} mu[d] * a[k/d] (inv zeta, a \ast mu)
template
<class T> vector<T> zeta(vector<T> a, bool inv = false) {
    int n = a.size() - 1;
    for (int p : primes)
        if (inv) for (int i = n / p; i >= 1; i--) a[i * p] -= a[i];
        else for (int i = 1; i <= n / p; i++) a[i * p] += a[i];
    return a;
}
// less implementation
if (inv) for (int i = 1; i <= n; i++)
    for (int j = 2 * i; j <= n; j += i) a[j] -= a[i];

```

```

    else for (int i = n; i >= 1; i--)
        for (int j = 2 * i; j <= n; j += i) a[j] += a[i];
}
// ck = sum_{lcm(i,j)=k} a[i] * b[j]
template<class T> vector<T> conv(vector<T> a, vector<T> b) {
    a = zeta(a), b = zeta(b);
    for (int i = 1; i < a.size(); i++) a[i] *= b[i];
    return zeta(a, true);
}

```

6.13 exLucas [f8047f]

```

ll legendre(ll n, int p) { // n! 中質數 p 的幕次
    ll res = 0;
    while (n) { n /= p, res += n; }
    return res;
}
Z C_pe(ll n, ll m, int p, int e) {
    int pe = power(p, e);
    Z::setMod(pe); // inv 要用 exgcd
    vector<Z> fac(pe); fac[0] = 1;
    for (int j
        = 1; j < pe; j++) fac[j] = fac[j - 1] * (j % p ? j : 1);
    Z wilson =
        p == 2 && e >= 3 ? 1 : -1; // (p^e)! = wilson (mod p^e)
    function<Z(ll)> fastFac = [&](ll n)
    { return n ? power
        (wilson, n / pe) * fac[n % pe] * fastFac(n / p) : 1; };
    ll vp = legendre(n, p) - legendre(m, p) - legendre(n - m, p);
    if (vp >= e) return 0;
    return power
        (Z(p), vp) * fastFac(n) / (fastFac(m) * fastFac(n - m));
}
ll exlucas(ll n, ll m, int mod) {
    vector<pair<int, int>> a;
    for (int i = 2; i * i <= mod; i++) {
        if (mod % i) continue;
        int e = 0, p = 1;
        while (mod % i == 0) mod /= i, e++, p *= i;
        a.emplace_back(C_pe(n, m, i, e).x, p);
    }
    if (mod != 1) a.emplace_back(C_pe(n, m, mod, 1).x, mod);
    return CRT(a);
}

```

6.14 Quadratic Residue [db1f2f]

```

int jacobi(int x, int p) {
    int s = 1;
    for (; p > 1; ) {
        x %= p;
        if (x == 0) return 0;
        const int r = __builtin_ctz(x);
        if ((r & 1) && ((p + 2) & 4)) s = -s;
        x >>= r;
        if (x & p & 2) s = -s;
        swap(x, p);
    }
    return s;
}
template<class Z> int quadraticResidue(Z x) {
    int p = Z::p();
    if (p == 2) return x.x & 1;
    const int jc = jacobi(x.x, p);
    if (jc == 0) return 0;
    if (jc == -1) return -1;
    Z b, d;
    while (true) {
        b = rand(), d = b * b - x;
        if (jacobi(d.x, p) == -1) break;
    }
    Z f0 = b, f1 = 1, g0 = 1, g1 = 0, tmp;
    for (int e = (p + 1) >> 1; e; e >>= 1) {
        if (e & 1) {
            tmp = g0 * f0 + d * (g1 * f1);
            g1 = g0 * f1 + g1 * f0, g0 = tmp;
        }
        tmp = f0 * f0 + d * (f1 * f1);
        f1 = f0 * f1 * 2, f0 = tmp;
    }
    return min(g0.x, p - g0.x);
}

```

6.15 BSGS [ed42da]

```

// a^x = b (mod m)
// x = A * sq - B (0 <= A, B <= sq)
vector<int> BSGS(int a, int b, int m) { // gcd(a, m) = 1
    Z::setMod(m);
    unordered_map<int, int> mp;
    ll sq = 1; while (sq * sq < m) sq++;
    Z rhs = b; vector<int> res;
    for (int B = 0; B <= sq; B++) {
        mp[rhs.x] = B;
        rhs *= a;
    }
    Z a_pow_sq = power(Z(a), sq), lhs = a_pow_sq;
    for (int A = 1; A <= sq; A++) lhs *= a_pow_sq;
    if (mp.find(lhs.x) != mp.end()) {
        int B = mp[lhs.x];
        res.push_back(A * sq - B);
    }
}

```

```

    return res;
}
int exBSGS(int a, int b, int m) {
    if (b == 1 || m == 1) return 0;
    if (!a) return b ? -1 : 1;
    int k = 0;
    ll tmp = 1;
    while (__gcd(a, m) != 1) {
        int d = __gcd(a, m);
        if (b % d != 0) return -1;
        b /= d, m /= d, k++;
        tmp = tmp * (a / d) % m;
        if (tmp == b) return k;
    }
    Z::setMod(m);
    Z b_ = Z(b) * Z(tmp).inv();
    auto xs = BSGS(a, b_.x, m);
    return xs
        .empty() ? -1 : *min_element(xs.begin(), xs.end()) + k;
}

```

7 Search and Greedy

7.1 Binary Search [d41d8c]

```

void binarySearch() {
    // 二分找上界
    // 如果無解會 = 原 lo, lo 要先 - 1
    while (lo < hi) {
        int x = (lo + hi + 1) / 2;
        if (check(x)) lo = x;
        else hi = x - 1;
    }
    cout << lo;
    // 二分找下界
    // 如果無解會 = 原 hi, hi 要先 + 1
    while (lo < hi) {
        int x = (lo + hi) / 2;
        if (check(x)) hi = x;
        else lo = x + 1;
    }
    cout << lo;
}

```

7.2 Ternary Search [d41d8c]

```

void ternarySearch() {
    int lo = 0, hi = 10;
    while (lo < hi) {
        int xl = lo + (hi - lo) / 3;
        int xr = hi - (hi - lo) / 3;
        int resl = calc(xl), resr = calc(xr);
        if (resl < resr) {
            lo = xl + 1;
        } else {
            hi = xr - 1;
        }
    }
}

```

8 DP

8.1 Projects [154da3]

```

void projects() { // 排程有權重問題，輸出價值最多且時間最少
    int n; cin >> n;
    vector<array<int, 4>> a(n + 1);
    for (int i = 1; i <= n; i++) {
        int u, v, w; cin >> u >> v >> w;
        a[i] = {v, u, w, i};
    }
    sort(a.begin(), a.end());
    vector<array<ll, 4>> dp(n + 1);
    // {w, time, 有沒選, 上個是誰}
    vector<ll> vs(n + 1);
    for (int i = 1; i <= n; i++) vs[i] = a[i][0];
    for (int i = 1; i <= n; i++) {
        int id = --lower_bound(all(vs), a[i][1]) - vs.begin();
        dp[i] = dp[id - 1];
        ll nw = dp[id][0] + a[i][2];
        ll nt = dp[id][1] + a[i][0] - a[i][1];
        if (dp[i][0] < nw || dp[i][0] == nw && dp[i][1] > nt) {
            dp[i] = {nw, nt, 1, id};
        }
    }
    vector<int> ans;
    for (int i = n; i != 0; ) {
        if (dp[i][2]) {
            ans.push_back(a[i][3]);
            i = dp[i][1];
        } else i--;
    } reverse(ans.begin(), ans.end());
}

```

8.2 Bitmask [eedcfc]

```

void hamiltonianPath() {
    int n, m; cin >> n >> m;
    vector<vector<int>> adj(n);
    for (int i = 0; i < m; i++) {

```

```

    int u, v; cin >> u >> v;
    adj[--v].push_back(--u);
}
// 以...為終點，走過...
vector dp(n, vector<int>(1 << n));
dp[0][1] = 1;
for (int mask = 1; mask < 1 << n; mask++) {
    if ((mask & 1) == 0) continue;
    for (int i = 0; i < n; i++) {
        if ((mask >> i & 1) == 0) continue;
        if (i == n - 1 && mask != (1 << n) - 1) continue;
        int pre = mask ^ (1 << i);
        for (int j : adj[i]) {
            if ((pre >> j & 1) == 0) continue;
            dp[i][mask] = (dp[i][mask] + dp[j][pre]) % Mod;
        }
    }
}
cout << dp[n - 1][(1 << n) - 1] << "\n";
}

void minClique() { // 移掉一些邊，讓整張圖由最少團組成
    int n, m;
    cin >> n >> m;
    vector<bitset<N>> g(n);
    for (int i = 0; i < m; i++) {
        int u, v; cin >> u >> v;
        u--; v--; g[u][v] = g[v][u] = 1;
    }
    vector<int> dp(1 << n, inf);
    dp[0] = 1;
    for (int mask = 0; mask < 1 << n; mask++) { // 先正常 dp
        for (int i = 0; i < n; i++) {
            if (mask & (1 << i)) {
                int pre = mask ^ (1 << i);
                if (dp[pre] == 1 && (g[i] & bitset<N>(pre)) == pre)
                    dp[mask] = 1; // i 有連到所有 pre
            }
        }
    }
    for (int mask = 0; mask < 1 << n; mask++) // 然後枚舉子集 dp
        for (int sub = mask; sub; --sub &= mask)
            dp[mask] = min(dp[mask], dp[sub] + dp[mask ^ sub]);
    cout << dp[(1 << n) - 1] << "\n";
}

```

8.3 Monotonic Queue [c9ba14]

```

// 應用: dp(i) = h(i) + max(A(j)), for l(i) <= j <= r(i)
// A(j) 可能包含 dp(j), h(i) 可 0(1)
void boundedKnapsack() {
    int n, k; // 0(nk)
    vector<int> w(n), v(n), num(n);
    deque<int> q;
    // 將同餘的數分在同一組
    // 每次取出連續 num[i] 格中最大值
    // g_x = max_{k=0}^{num[i]} (g_{x-k} + v_i * k)
    // G_x = g_{x-k} - v_i * k
    // x 代 x-k => v_i * (x-k)
    // g_x = max_{k=0}^{num[i]} (G_{x-k} + v_i * x)
    vector<vector<ll>> dp(2, vector<ll>(k + 1));
    for (int i = 0; i < n; i++) {
        for (int r = 0; r < w[i]; r++) { // 餘數
            q.clear(); // q 記錄在 x = i 時的 dp 有單調性
            for (int x = 0; x * w[i] + r <= k; x++) {
                while (!q.empty() &&
                    q.front() < x - num[i]) q.pop_front(); // 維護遞減
                ll nxt = dp[0][x * w[i] + r] - x * v[i];
                while (!q.empty() && dp[0][q.back()]
                    * w[i] + r - q.back() * v[i] < nxt) q.pop_back();
                q.push_back(x);
                dp[1][x * w[i] + r] = dp[0][q.
                    front() * w[i] + r] - q.front() * v[i] + x * v[i];
            }
        }
        swap(dp[0], dp[1]);
    }
    cout << dp[0][k] << "\n";
}

```

8.4 Digit [1c2af1]

```

// 只含 0, 1, 8 的回文數
const int N = 44;
ll memo[N + 1][N + 1]; // memset -1, once
ll solve(string s) {
    vector<int> a {0};
    while (!s.empty()) {
        a.push_back(s.back() - '0');
        s.pop_back();
    }
    int len = a.size() - 1;
    vector<int> now(len + 1, -1); // 紀錄目前的數字
    return [&](this
        auto &&self, int p, int eff, bool f0, bool lim) -> ll {
        if (!p) return 1; // eff: 扣除前導 0 的長度
        if (!lim && !f0 && memo[p][eff] != -1) return memo[p][eff];
        ll res = 0;
        int lst = lim ? a[p] : 9; // or 1 for binary
        for (int i = 0; i <= lst; i++) {

```

```

            if (i != 0 && i != 1 && i != 8) continue;
            bool nlim = lim && i == lst;
            now[p] = i;
            if (f0 && i == 0) { // 處理前導零
                res += self(p - 1, eff - 1, true, nlim);
            } else if (p > eff / 2) { // 前半段
                res += self(p - 1, eff, false, nlim);
            } else if (now[p] == now[eff - p + 1]) {
                res += self(p - 1, eff, false, nlim);
            }
        }
        if (!lim && !f0) memo[p][eff] = res;
        return res;
    } (len, len, true, true);
}
ll memo[N + 1][10]; // memset -1, once
// 不貼上限、
    不前導零、枚舉 p 往後所有數字中，每個 digit x 出現的次數
ll solve(int x, ll n) {
    vector<int> a {0};
    vector<ll> sum {0}; // 貼上限時，位數 p 以下有多少數字
    while (n) {
        a.push_back(n % 10);
        sum.push_back
            (sum.back() + (n % 10) * power(10, a.size() - 2));
        n /= 10;
    }
    int len = a.size() - 1;
    return [&](this auto &&self, int p, bool f0, bool lim) -> ll {
        if (!p) return 0; // 貢獻來自於當前 digit
        if (!lim && !f0 && memo[p][x] != -1) return memo[p][x];
        ll res = 0;
        int lst = lim ? a[p] : 9; // or 1 for binary
        for (int i = 0; i <= lst; i++) {
            bool nlim = lim && i == lst;
            if (f0 && i == 0) { // 處理前導零
                res += self(p - 1, true, nlim);
            } else if (nlim) { // 枚舉的前幾位都貼在 n 上
                res += self(p - 1, false, nlim);
                if (i == x) res += sum[p - 1] + 1;
            } else {
                res += self(p - 1, false, nlim);
                if (i == x) res += power(10, p - 1);
            }
        }
        if (!lim && !f0) memo[p][x] = res;
        return res;
    } (len, true, true);
}
// 連續兩位 digit > 1
ll memo[N + 1][10]; // memset -1, once
// 不貼上限、
    不前導零、前一位為 prv 時、枚舉 p 往後所有數字的合法個數
ll solve(ll x) { // 建 a...
    return [&](this
        auto &&self, int p, int prv, bool f0, bool lim) -> ll {
        if (!p) return 1; // 貢獻來自於當前數字數量
        if (!lim && !f0 && memo[p][prv] != -1) return memo[p][prv];
        ll res = 0;
        int lst = lim ? a[p] : 9; // or 1 for binary
        for (int i = 0; i <= lst; i++) {
            if (prv != -1 && abs(i - prv) < 2) continue;
            bool nlim = lim && i == lst;
            if (f0 && i == 0) { // 處理前導零
                res += self(p - 1, -1, true, nlim);
            } else {
                res += self(p - 1, i, false, nlim);
            }
        }
        if (!lim && !f0) memo[p][prv] = res;
        return res;
    } (len, -1, true, true);
}

```

8.5 SOS [ec9d5a]

```

// 使用情況: 跟 bit 與(被)包含有關，且 x 在 1E6 左右
// 題目: 一數組，問有多少所有數 & 起來為 0 的集合數
// dp[
    x] 代表包含 x 的 y 個數(比 x 大且 bit 1 全包含 x 的有幾個)
// 答案應該包含在 dp[0] 內，但是有重複元素，所以考慮容斥
// => ans = \sum_{i=0}^n (-1)^{pop_count(i)} 2^{dp[i]-1}
// => 全
    部為 0 的個數 - 至少一個為 1 的個數 + 至少兩個為 1 的個數
void solve(int n, vector<int> a) {
    int m = __lg(*max_element(a.begin(), a.end())) + 1;
    // 定義 dp[mask] 為 mask 被包含於 a[i] 的 i 個數
    vector<Z> dp(1 << m);
    for (int i = 0; i < n; i++) dp[a[i]] += 1;
    for (int i = 0; i < m; i++) {
        for (int mask = 0; mask < 1 << m; mask++) {
            if (mask >> i & 1) {
                int pre = mask ^ (1 << i);
                dp[pre] += dp[mask];
            }
        }
    }
}

```

```

for (int i = 0; i < 1 << m; i++) {
    dp[i] = power(Z(2), int(dp[i])) - 1;
}
// 排容，dp[i]: & 起來為 i 的集合數
for (int i = 0; i < m; i++) {
    for (int mask = 0; mask < 1 << m; mask++) {
        if (mask >> i & 1) {
            int pre = mask ^ (1 << i);
            dp[pre] -= dp[mask];
        }
    }
}
// x | y = x, 代表包含於 x 的 y 個數，定義為 dp[x][0]
// x & y = x, 代表包含 x 的 y 個數，定義為 dp[x][1]
// x & y
// != 0, 代表至少有一個位元都為 1 的 y 個數，= n - dp[~x][0]
void solve(int n, vector<int> a) {
    int m = __lg(*max_element(a.begin(), a.end())) + 1;
    vector<array<ll, 2>> dp(1 << m);
    for (int i = 0; i < n; i++) dp[a[i]][0]++, dp[a[i]][1]++;
    for (int i = 0; i < m; i++) {
        for (int mask = 0; mask < 1 << m; mask++) {
            if (mask >> i & 1) {
                int pre = mask ^ (1 << i);
                dp[mask][0] += dp[pre][0];
                dp[pre][1] += dp[mask][1];
            }
        }
    }
}

```

8.6 CHT [bdb950]

```

// 應用: dp(x) = C(x) + min/max(A(i) * x + B(i)), for i < x
// 斜率遞減，求 min, 上凸包，自行轉化問題
// ex: m 遞增取 max => max(mx + b) = -min(-m x - b)
struct Line { // x 盡量從 1 開始
    ll m, b;
    Line(ll m = 0, ll b = 0) : m(m), b(b) {}
    ll eval(ll x) { return m * x + b; }
};
struct CHT { // 斜率單調
    deque<Line> hull;
    CHT() {}
    CHT(Line init) { addLine(init); }
    bool frontBad(Line &l1, Line &l2, ll x) {
        return l1.eval(x) >= l2.eval(x);
    }
    bool backBad(Line &l1, Line &l2, Line &l) {
        // 只要
        // l2 跟 l 的 x 交點 <= l1 跟 l 的 x 交點，l2 就用不到了
        return (l.b - l2.b) * (l1.m - l.m) <= (l.b - l1.b) * (l2.m - l.m);
    }
    void addLine(Line l) {
        if (hull.size() > 0 && hull.back().m == l.m) {
            if (hull.back().b <= l.b) return;
            else hull.pop_back();
        }
        while (hull.size() > 1 && backBad(hull[hull.size()-2], hull.back(), l)) hull.pop_back();
        hull.push_back(l);
    }
    ll query(ll x) { // 查詢沒單調性需要二分搜
        while (hull.size() > 0 && frontBad(hull[0], hull[1], x)) hull.pop_front();
        return hull[0].eval(x);
    }
};

```

8.7 DNC [a29240]

```

// 應用: 切 k 段問題，且滿足四邊形不等式
// w(a,c) + w(b,d) <= w(a,d) + w(b,c)
// dp[k][j] = min(dp[k-1][i] + cost[i][j])
// O(K N logN)
auto getCost = [&](int l, int r) {} // (l, r], 1-based
vector<ll> dp(n+1, inf); dp[0] = 0;
for (int j = 0; j < k; j++) {
    vector<ll> ndp(n+1, inf);
    [&](this auto &&self, int l, int r, int optl, int optr) {
        if (l > r) return;
        int m = (l + r) / 2, opt = 0;
        ndp[m] = inf;
        for (int i = optl; i <= min(m-1, optr); i++) {
            ll x = dp[i] + getCost(i, m);
            if (x < ndp[m]) ndp[m] = x, opt = i;
        }
        self(l, m-1, optl, opt);
        self(m+1, r, opt, optr);
    } (1, n, j, n-1);
    dp = ndp;
}

```

8.8 LiChao Segment Tree [2a9325]

```

// 應用: dp(i) = h(i) + min/max(A(j)X(i) + B(j)), for j <= r(i)
// y = c + m x + b

```

```

template<class T, class F = less<ll>>
struct LiChaoSeg {
    F cmp = F();
    static const T inf = max(numeric_limits<T>::lowest() / 2, numeric_limits<T>::max() / 2, F());
    struct Line {
        T m, b;
        Line(T m = 0, T b = inf) : m(m), b(b) {}
        T eval(T x) const { return m * x + b; }
    };
    struct Node {
        Line line;
        ll l = -1, r = -1;
    };
    ll n;
    vector<Node> nd;
    LiChaoSeg(ll n) : n(n) { newNode(); }
    void addLine(Line line) { update(0, 0, n, line); }
    void rangeAddLine(Line line, ll ql, ll qr) { rangeUpdate(0, 0, n, ql, qr, line); }
    T query(ll x) { return query(x, 0, 0, n); }
private:
    int newNode() {
        nd.emplace_back();
        return nd.size() - 1;
    }
    void update(int p, ll l, ll r, Line line) {
        ll m = (l + r) / 2;
        bool left = cmp(line.eval(l), nd[p].line.eval(l));
        bool mid = cmp(line.eval(m), nd[p].line.eval(m));
        if (mid) swap(nd[p].line, line);
        if (r - l == 1) return;
        if (left != mid) {
            if (nd[p].l == -1) nd[p].l = newNode();
            update(nd[p].l, l, m, line);
        } else {
            if (nd[p].r == -1) nd[p].r = newNode();
            update(nd[p].r, m, r, line);
        }
    }
    void rangeUpdate(int p, ll l, ll r, ll ql, ll qr, Line line) {
        if (r <= ql || l >= qr) return;
        if (ql <= l && r <= qr) return update(p, l, r, line);
        if (nd[p].l == -1) nd[p].l = newNode();
        if (nd[p].r == -1) nd[p].r = newNode();
        ll m = (l + r) / 2;
        rangeUpdate(nd[p].l, l, m, ql, qr, line);
        rangeUpdate(nd[p].r, m, r, ql, qr, line);
    }
    T query(ll x, int p, ll l, ll r) {
        if (p == -1) return inf;
        ll m = (l + r) / 2;
        if (x < m) return min(nd[p].line.eval(x), query(x, nd[p].l, l, m), cmp);
        else return min(nd[p].line.eval(x), query(x, nd[p].r, m, r), cmp);
    }
};

```

9 Geometry

9.1 Basic [0f20c9]

```

const double eps = 1E-9;
template<class T> struct Pt {
    T x, y;
    Pt(T x = 0, T y = 0) : x(x), y(y) {}
    Pt operator-(Pt p) const { return Pt(-x, -y); }
    Pt operator+(Pt p) const { return Pt(x + p.x, y + p.y); }
    Pt operator-(Pt p) const { return Pt(x - p.x, y - p.y); }
    Pt operator*(T k) const { return Pt(x * k, y * k); }
    Pt operator/(T k) const { return Pt(x / k, y / k); }
    bool operator==(Pt p) const { return x == p.x && y == p.y; }
    bool operator!=(Pt p) const { return x != p.x || y != p.y; }
    friend istream &operator>>(istream &is, Pt &p) {
        return is >> p.x >> p.y;
    }
    friend ostream &operator<<(ostream &os, const Pt &p) {
        return os << "(" << p.x << ", " << p.y << ")";
    }
};
int sign(double x) {
    return fabs(x) <= eps ? 0 : (x > 0 ? 1 : -1);
}
using P = Pt<double>;

struct Line { P a, b; };
template<class T>
    T dot(Pt<T> a, Pt<T> b) { return a.x * b.x + a.y * b.y; }
template<class T> T
    cross(Pt<T> a, Pt<T> b) { return a.x * b.y - a.y * b.x; }
template<class T> T abs2(Pt<T> p) { return dot(p, p); }
double abs(P p) { return sqrt(abs2(p)); }
double dist(P a, P b) { return abs(a - b); }
double abs(Line l) { return abs(l.a - l.b); }
int dir(P p, Line l) // left -1, right 1, on 0
{ return -sign(cross(l.b - l.a, p - l.a)); }
bool btw(P p, Line l) // c on segment ab?
{ return dir(p, l) == 0 && sign(dot(p - l.a, p - l.b)) <= 0; }
P norm(P p) { return p / abs(p); }
P rot(P p) { return {-p.y, p.x}; } // 90 degree CCW
P rot(P p, double d) { // CCW, d = degree * PI / 180
    double c = cos(d), s = sin(d);
    return P(p.x * c - p.y * s, p.x * s + p.y * c);
}

```

```

}

bool parallel(Line l1, Line l2)
{ return sign(cross(l1.b - l1.a, l2.b - l2.a)) == 0; }
P lineIntersection(Line l1, Line l2)
{ return l1.a + (l1.b - l1.a) * (cross(l2.b - l2.a, l1.a - l1.b)) / cross(l2.b - l2.a, l1.a - l1.b)); }
bool pointOnSegment(P p, Line l)
{ return dir(p, l) == 0 && sign(dot(p - l.a, p - l.b)) <= 0; }
P projvec(P p, Line l) {
    P v = l.b - l.a;
    return l.a + v * (dot(p - l.a, v) / abs2(v));
}

// 0 : not intersect
// 1 : strictly intersect
// 2 : overlap
// 3 : intersect at endpoint
tuple<int, P, P> segmentIntersection(Line l1, Line l2) {
    if (sign(max(l1.a.x, l1.b.x) - min(l2.a.x, l2.b.x)) == -1 ||
        sign(min(l1.a.x, l1.b.x) - max(l2.a.x, l2.b.x)) == 1 ||
        sign(max(l1.a.y, l1.b.y) - min(l2.a.y, l2.b.y)) == -1 ||
        sign(min(l1.a.y, l1.b.y) - max(l2.a.y, l2.b.y)) == 1)
        return {0, {}, {}};
    if (parallel(l1, l2)) {
        if (dir(l1.a, l2) != 0) return {0, {}, {}};
        else {
            vector<P> res;
            if (pointOnSegment(l1.a, l2)) res.push_back(l1.a);
            if (pointOnSegment(l1.b, l2)) res.push_back(l1.b);
            if (pointOnSegment(l2.a, l1)) res.push_back(l2.a);
            if (pointOnSegment(l2.b, l1)) res.push_back(l2.b);
            if (res.empty()) return {0, {}, {}};
            sort(res.begin(), res.end(), [](const P &a, const P &b)
                { return sign(a.x - b.x) == 0 ? a.y < b.y : a.x < b.x; });
            res.resize(unique(res.begin(), res.end()) - res.begin());
            if (res.size() == 1) return {3, res[0], res[0]};
            else return {2, res.front(), res.back()};
        }
    }
    auto cp1 = dir(l2.b, l1), cp2 = dir(l2.a, l1), cp3 = dir(l1.b, l2), cp4 = dir(l1.a, l2);
    if ((cp1 > 0 && cp2 > 0) || (cp1 < 0 && cp2 < 0) || (cp3 > 0 && cp4 > 0) || (cp3 < 0 && cp4 < 0))
        return {0, {}, {}};
    P p = lineIntersection(l1, l2);
    if (cp1 != 0 && cp2 != 0 && cp3 != 0 && cp4 != 0) return {1, p, p};
    else return {3, p, p};
}

vector<P> convexHull(vector<P> a) {
    sort(a.begin(), a.end(), [](const P &a, const P &b)
        { return sign(a.x - b.x) == 0 ? a.y < b.y : a.x < b.x; });
    a.resize(unique(a.begin(), a.end()) - a.begin());
    if (a.size() <= 1) return a;
    vector<P> h(a.size() * 2);
    int s = 0, t = 0;
    for (int i = 0; i < 2; i++, s = -t) {
        for (P p : a) {
            while (t >= s + 2 && sign(cross(h[t - 1] - h[t - 2], p - h[t - 2])) <= 0) t--;
            h[t++] = p;
        }
        reverse(a.begin(), a.end());
    }
    return {h.begin(), h.begin() + t};
}

double distPL(P p, Line l)
{ return abs(cross(l.a - l.b, l.a - p)) / abs(l); }
double distancePS(P p, Line l) {
    if (sign(dot(p - l.a, l.b - l.a)) < 0) return dist(p, l.a);
    if (sign(dot(p - l.b, l.a - l.b)) < 0) return dist(p, l.b);
    return distPL(p, l);
}

double distanceSS(Line l1, Line l2) {
    if (get<0>(segmentIntersection(l1, l2)) != 0) return 0.0;
    return min({distancePS(l1.a, l2), distancePS(l1.b, l2), distancePS(l2.a, l1), distancePS(l2.b, l1)});
}

bool lineIntersectsPolygon(Line l, const vector<P> &p) {
    int n = p.size();
    for (int i = 0; i < n; i++) {
        P a = p[i], b = p[(i + 1) % n];
        if (dir(a, l) == 0 || dir(b, l) == 0) return true;
        if ((dir(a, l) < 0) ^ (dir(b, l) < 0)) return true;
    }
    return false;
}

bool pointInPolygon(P a, const vector<P> &p) {
    int n = p.size(), t = 0;
    for (int i = 0; i < n; i++)
        if (pointOnSegment(a, {p[i], p[(i + 1) % n]})) return true;
    for (int i = 0; i < n; i++) {
        P u = p[i], v = p[(i + 1) % n];
        if (u.x < a.x && v.x >= a.x && dir(a, {v, u}) < 0) t ^= 1;
        if (u.x >= a.x && v.x < a.x && dir(a, {u, v}) < 0) t ^= 1;
    }
    return t == 1;
}

```

```

}
// 0 : strictly outside
// 1 : on boundary
// 2 : strictly inside
int pointInConvexPolygon(P a, const vector<P> &p) {
    int n = p.size();
    if (n == 0) return 0;
    else if (n <= 2) return pointOnSegment(a, {p[0], p.back()});
    if (pointOnSegment(a, {p[0], p[1]})) return 1;
    if (pointOnSegment(a, {p[0], p[n - 1]})) return 1;
    else if (dir(a, {p[0], p[1]}) < 0 || dir(a, {p[0], p[n - 1]}) < 0) return 0;
    int lo = 1, hi = n - 2;
    while (lo < hi) {
        int x = (lo + hi + 1) / 2;
        if (dir(a, {p[0], p[x]}) < 0) lo = x;
        else hi = x - 1;
    }
    if (dir(a, {p[lo], p[lo + 1]}) < 0) return 2;
    else return pointOnSegment(a, {p[lo], p[lo + 1]});
}

bool segmentInPolygon(Line l, const vector<P> &p) {
    int n = p.size();
    vector<P> a {l.a, l.b};
    for (int i = 0; i < n; i++) {
        auto [t, p1, p2] = segmentIntersection(l, {p[i], p[(i + 1) % n]});
        if (t == 1 || t == 3) a.push_back(p1);
        else if (t == 2) a.push_back(p1), a.push_back(p2);
    }
    sort(a.begin(), a.end(), [](const P &a, const P &b)
        { return sign(a.x - b.x) == 0 ? a.y < b.y : a.x < b.x; });
    for (int i = 0; i + 1 < a.size(); i++) {
        if (sign(dist(a[i], a[i + 1])) == 0) continue;
        if (!pointInPolygon((a[i] + a[i + 1]) / 2, p)) return false;
    }
    return true;
}

// non-strict
-> >= 都改 >, assert 拿掉, 在外面特判直線(area == 0)
vector<P> hp(vector<Line> lines) {
    auto sgn = [](P p) {
        return p.y > 0 || (p.y == 0 && p.x > 0) ? 1 : -1;
    };
    sort(lines.begin(), lines.end(), [&](auto l1, auto l2) {
        auto d1 = l1.b - l1.a;
        auto d2 = l2.b - l2.a;
        if (sgn(d1) != sgn(d2)) return sgn(d1) == 1;
        return cross(d1, d2) > 0;
    });
    deque<Line> ls;
    deque<P> ps;
    for (auto l : lines) {
        if (ls.empty()) {
            ls.push_back(l);
            continue;
        }
        while (!ps.empty() && dir(ps.back(), l) >= 0) ps.pop_back(), ls.pop_back();
        while (!ps.empty() && dir(ps[0], l) >= 0) ps.pop_front(), ls.pop_front();
        if (sign(cross(l.b - l.a, ls.back().b - ls.back().a)) == 0) { // 平行
            if (sign(dot(l.b - l.a, ls.back().b - ls.back().a)) > 0) { // 同向
                if (dir(ls.back().a, l) >= 0) { // l 在內側
                    assert(ls.size() == 1);
                    ls[0] = l;
                }
                continue;
            }
            return {};
        }
        ps.push_back(lineIntersection(ls.back(), l));
        ls.push_back(l);
    }
    while (!ps.empty() && dir(ps.back(), ls[0]) >= 0)
        ps.pop_back(), ls.pop_back();
    if (ls.size() <= 2) return {};
    ps.push_back(lineIntersection(ls[0], ls.back()));
    return vector(ps.begin(), ps.end());
}

```

9.2 Min Euclidean Distance [2e8f35]

```

void solve(int n, vector<P> &a) { // recursive solution
    sort(a.begin(), a.end(), [](const P &a, const P &b)
        { return a.x == b.x ? a.y < b.y : a.x < b.x; });
    vector<P> t(n);
    cout << [&](this auto &&self, int l, int r) -> ll {
        if (l == r) return 9E18;
        int p = 0, m = (l + r) / 2;
        ll mid = a[m].x;
        ll ans = min(self(l, m), self(m + 1, r));
        merge(a.begin() + l, a.begin() + m + 1,
            a.begin() + m + 1, a.begin() + r + 1,
            t.begin(),
            [&](const P &a, const P &b) { return a.y < b.y; });
        copy(t.begin(), t.begin() + r - l + 1, a.begin() + l);
        for (int i = l; i <= r; i++)
            if ((mid - a[i].x) * (mid - a[i].x) <= ans) t[p++] = a[i];
    };
}

```



```

    for (int i = 0; i < p; i++) {
        for (int j = i + 1; j < p; j++) {
            if ((t[i].y - t[j].y) * (t[i].y - t[j].y) > ans) break;
            ans = min(ans, abs2(t[i] - t[j]));
        }
    }
    return ans;
} (0, n - 1) << "\n";
// K-D tree solution
struct Info {
    static constexpr int DIM = 2;
    array<ll, DIM> x, l, r;
    ll distl, distr;
    ll f(const Info &i) {
        ll ret = 0;
        if (i.L[0] > x[0]) ret += (i.L[0] - x[0]) * (i.L[0] - x[0]);
        if (i.R[0] < x[0]) ret += (x[0] - i.R[0]) * (x[0] - i.R[0]);
        if (i.L[1] > x[1]) ret += (i.L[1] - x[1]) * (i.L[1] - x[1]);
        if (i.R[1] < x[1]) ret += (x[1] - i.R[1]) * (x[1] - i.R[1]);
        return ret;
    }
    void pull(const Info &l, const Info &r) {
        distl = f(l), distr = f(r);
    }
};

struct KDTree { // 1-indexed
    static constexpr int DIM = Info::DIM;
    int n, rt;
    vector<Info> info;
    vector<int> l, r;
    KDTree(const vector<Info> &
            info) : n(info.size()), info(info), l(n + 1), r(n + 1) {
        rt = build(1, n);
    }
    void pull(int p) {
        info[p].L = info[p].R = info[p].x;
        info[p].pull(info[l[p]], info[r[p]]);
        for (int ch : {l[p], r[p]}) {
            if (!ch) continue;
            for (int k = 0; k < DIM; k++) {
                info[p].L[k] = min(info[p].L[k], info[ch].L[k]);
                info[p].R[k] = max(info[p].R[k], info[ch].R[k]);
            }
        }
    }
    int build(int l, int r) {
        if (r == l) return 0;
        int m = (l + r) / 2;
        array<double, DIM> av = {}, va = {};
        for (int i = l; i < r; i++)
            for (int d = 0; d < DIM; d++)
                av[d] += info[i].x[d];
        for (int d = 0; d < DIM; d++)
            av[d] /= (double)(r - l);
        for (int i = l; i < r; i++)
            for (int d = 0; d < DIM; d++)
                va[d] +=
                    (info[i].x[d] - av[d]) * (info[i].x[d] - av[d]);
        int dep = max_element(va.begin(), va.end()) - va.begin();
        nth_element
            (info.begin() + l, info.begin() + m, info.begin() + r,
             [&](const Info
                 &x, const Info &y) { return x.x[dep] < y.x[dep]; });
        this->l[m] = build(l, m);
        this->r[m] = build(m + 1, r);
        pull(m); return m;
    }
    ll ans = 9E18;
    ll dist(int a, int b) {
        return (info[a].
                x[0] - info[b].x[0]) * (info[a].x[0] - info[b].x[0]) +
                (info[a].
                x[1] - info[b].x[1]) * (info[a].x[1] - info[b].x[1]);
    }
    void query(int p, int x) {
        if (!p) return;
        if (p != x) ans = min(ans, dist(x, p));
        ll distl = info[x].f(info[l[p]]);
        ll distr = info[x].f(info[r[p]]);
        if (distl < ans && distr < ans) {
            if (distl < distr) {
                query(l[p], x);
                if (distr < ans) query(r[p], x);
            } else {
                query(r[p], x);
                if (distl < ans) query(l[p], x);
            }
        } else {
            if (distl < ans) query(l[p], x);
            if (distr < ans) query(r[p], x);
        }
    }
};

```

9.3 Max Euclidean Distance [7e3b52]

```
tuple<ll, int, int> maxEuclideanDistance(vector<P> a) {
```

```

    auto get = [&](P p, Line l) -> ll
    { return abs(cross(l.a - l.b, l.a - p)); };
    ll res = 0; int n = a.size(), x, y, id = 2;
    a.push_back(a.front());
    if (n == 1) return {0, 0, 0};
    if (n == 2) return {abs2(a[0] - a[1]), 0, 1};
    for (int i = 0; i < n; i++) {
        while (get(a[id], {a[i],
                           a[i + 1]}) <= get(a[(id + 1) % n], {a[i], a[i + 1]}))
            id = (id + 1) % n;
        if (res < abs2(a[i] - a[id])) {
            res = abs2(a[i] - a[id]);
            x = i, y = id;
        }
        if (res < abs2(a[i + 1] - a[id])) {
            res = abs2(a[i + 1] - a[id]);
            x = i + 1, y = id;
        }
    }
    return {res, x % n, y};
}

```

9.4 Min Circle Cover [96a4b7]

```

pair<double, P> minCircleCover(int n, vector<P> a) {
    shuffle(a.begin(), a.end(), rng);
    P c = a[0]; double r = 0;
    for (int i = 1; i < n; i++) if (sign(abs(c - a[i]) - r) > 0) {
        c = a[i], r = 0;
        for (int j = 0; j < i; j++) if (sign(abs(c - a[j]) - r) > 0) {
            c = (a[i] + a[j]) / 2.0;
            r = abs(c - a[i]);
            for (int k = 0; k < j; k++) if (sign(abs(c - a[k]) - r) > 0) {
                P p = (a[j] + a[i]) / 2;
                P q = (a[j] + a[k]) / 2;
                if (parallel({a[i], a[j]}, {a[j], a[k]})) continue;
                c = lineIntersection({p,
                                     p + rot(a[j] - a[i])}, {q, q + rot(a[k] - a[j])});
                r = abs(c - a[i]);
            }
        }
    }
    return {r, c};
}

```

9.5 Min Rectangle Cover [3a0f29]

```

pair<double, vector<P>> minRectangleCover(vector<P> p) {
    if (p.size() <= 2) return {0, {}};
    auto get = [&](P p, Line l) -> double {
        return abs(cross(l.a - l.b, l.a - p));
    }; // line 到 p 圍成的四邊形面積
    int n = p.size(), j = 2, l = 1, r = 1;
    p.push_back(p.front());
    double ans = 8E18;
    vector<P> ps;
    for (int i = 0; i < n; i++) {
        while (get(p[j], {p[i],
                           p[i + 1]}) <= get(p[(j + 1) % n], {p[i], p[i + 1]}))
            j = (j + 1) % n;
        while (dot(p[i + 1] - p[i], p[r] - p[i]) <= dot(p[i + 1] - p[i], p[(r + 1) % n] - p[i]))
            r = (r + 1) % n;
        if (i == 0) l = j;
        while (dot(p[i + 1] - p[i], p[l] - p[i]) >= dot(p[i + 1] - p[i], p[(l + 1) % n] - p[i]))
            l = (l + 1) % n;
        double area = get(p[j], {p[i], p[i + 1]});
        double w = dot(p[i] - p[i + 1],
                       p[l] - p[i]) + dot(p[i + 1] - p[i], p[r] - p[i]);
        area *= w / abs2(p[i + 1] - p[i]);
        if (area < ans) {
            ps.clear(), ans = area;
            Line l1 {p[i], p[i + 1]};
            for (auto u : {p[r], p[j], p[l], p[i]}) {
                if (u == l1.b) {
                    ps.push_back(u);
                    l1 = {u, u + rot(l1.b - l1.a)};
                } else {
                    Line l2 = {u, u + rot(l1.b - l1.a)};
                    P res = lineIntersection(l1, l2);
                    ps.push_back(res);
                    l1 = {res, u};
                }
            }
        }
    }
    return {ans, ps};
}

```

9.6 Convex Polygon Distance [92874d]

```

// CCW needed, O(n + m)
double convexPolyDist(vector<P> a, vector<P> b) {
    int n = a.size(), m = b.size();
    int sa = 0, sb = 0;
    for (int i = 1; i < n; i++) if (a[i].y < a[sa].y) sa = i;
    for (int i = 1; i < m; i++) if (b[i].y > b[sb].y) sb = i;
    double ans = 1E18;

```



```

for (int i = 0; i < n; i++) {
    P na = a[sa], nb = a[(sa + 1) % n];
    P ea = nb - na;
    while (sign(cross(ea, b[(sb + 1) % m] - b[sb])) > 0)
        sb = (sb + 1) % m;
    chmin(ans, distanceSS({na, nb}, {b[sb], b[(sb + 1) % m]}));
    sa = (sa + 1) % n;
}
return ans;
}

```

9.7 Lattice Points [fbecbc]

```

// (interior, boundary)
pair<ll, ll> latticePoints(int n, vector<P> p, ll area2) {
    ll bnd = 0;
    p.push_back(p[0]);
    for (int i = 0; i < n; i++)
        bnd += gcd(abs(p[i + 1].x - p[i].x), abs(p[i + 1].y - p[i].y));
    return {(area2 - bnd) / 2 + 1, bnd};
}

```

9.8 Polygon Union Area [24443d]

```

double polygonUnion(vector<vector<P>> ps) { // CCW needed
    int n = ps.size();
    for (auto &v : ps) v.push_back(v[0]);
    double res = 0;
    auto seg = [&](P o, P a, P b) -> double {
        if (b.x - a.x == 0) return (o.y - a.y) / (b.y - a.y);
        return (o.x - a.x) / (b.x - a.x);
    };
    for (int pi = 0; pi < n; pi++) {
        for (int i = 0; i + 1 < ps[pi].size(); i++) {
            vector<pair<double, int>> e {{0, 0}, {1, 0}};
            for (int pj = 0; pj < n; pj++) {
                if (pi == pj) continue;
                for (int j = 0; j + 1 < ps[pj].size(); j++) {
                    auto c1 = cross(ps[pi][i + 1] - ps[pi][i], ps[pj][j] - ps[pi][i]);
                    auto c2 = cross(ps[pi][i + 1] - ps[pi][i], ps[pj][j + 1] - ps[pi][i]);
                    if (c1 == 0 && c2 == 0) {
                        if (dot(ps[pi][i + 1] - ps[pi][i], ps[pj][j + 1] - ps[pj][j]) > 0 && (pi - pj) > 0) {
                            e.emplace_back(seg(ps[pj][j], ps[pi][i], ps[pi][i + 1]), 1);
                            e.emplace_back(seg(ps[pj][j + 1], ps[pi][i + 1], ps[pi][i]), -1);
                        }
                    } else {
                        auto s1 = cross(ps[pj][j + 1] - ps[pj][j], ps[pi][i] - ps[pj][j]);
                        auto s2 = cross(ps[pj][j] - ps[pj][j + 1], ps[pi][i + 1] - ps[pj][j]);
                        if (c1 >= 0 && c2 < 0) e.emplace_back(s1 / (s1 - s2), 1);
                        else if (c1 < 0 && c2 >= 0) e.emplace_back(s1 / (s1 - s2), -1);
                    }
                }
            }
            sort(e.begin(), e.end());
            double pre = clamp(e[0].first, 0.0, 1.0), sum = 0;
            int cov = e[0].second;
            for (int j = 1; j < e.size(); j++) {
                double now = clamp(e[j].first, 0.0, 1.0);
                if (!cov) sum += now - pre;
                cov += e[j].second;
                pre = now;
            }
            res += cross(ps[pi][i], ps[pi][i + 1]) * sum;
        }
    }
    return res / 2;
}

```

10 Polynomial

10.1 FFT [ce3602]

```

const double PI = acos(-1.0);
using cd = complex<double>;
vector<int> rev;
void fft(vector<cd> &a, bool inv = false) {
    int n = a.size();
    if (rev.size() != n) {
        int k = __builtin_ctz(n) - 1;
        rev.resize(n);
        for (int i = 0; i < n; i++)
            rev[i] = rev[i >> 1] >> 1 | (i & 1) << k;
    }
    for (int i = 0; i < n; i++)
        if (rev[i] < i) swap(a[i], a[rev[i]]);
    for (int k = 1; k < n; k *= 2) {
        double ang = (inv ? -1 : 1) * PI / k;
        cd wn(cos(ang), sin(ang));
        for (int i = 0; i < n; i += 2 * k) {
            cd w(1);

```

```

            for (int j = 0; j < k; j++, w = w * wn) {
                cd u = a[i + j];
                cd v = a[i + j + k] * w;
                a[i + j] = u + v;
                a[i + j + k] = u - v;
            }
        }
    }
    if (inv) for (auto &x : a) x /= n;
}
// i + j = k, res[1] = a[1] * b[0] + a[0] * b[1]
// reverse b for i - j = k, res[k + b.size() - 1]
template
<class T> vector<double> conv(vector<T> a, vector<T> b) {
    vector<cd> fa(a.begin(), a.end()), fb(b.begin(), b.end());
    int n = 1, tot = a.size() + b.size() - 1;
    while (n < tot) n <= 1;
    fa.resize(n), fb.resize(n);
    fft(fa), fft(fb);
    for (int i = 0; i < n; i++) fa[i] = fa[i] * fb[i];
    fft(fa, true);
    vector<double> res(tot);
    for (int i = 0; i < tot; i++) res[i] = fa[i].real();
    return res; // use lround to int
}

```

10.2 NTT [bb12cf]

```

template<class Z>
void ntt(vector<Z> &a, const int G = 3, bool inv = false) {
    static vector<int> rev;
    static vector<Z> w {0, 1};
    int n = a.size();
    if (rev.size() != n) {
        int k = __builtin_ctz(n) - 1;
        rev.resize(n);
        for (int i = 0; i < n; i++) rev[i] = rev[i >> 1] >> 1 | (i & 1) << k;
    }
    for (int i = 0; i < n; i++) if (rev[i] < i) swap(a[i], a[rev[i]]);
    if (w.size() < n) {
        int k = __builtin_ctz(w.size());
        w.resize(n);
        while ((1 << k) < n) {
            Z u = power(Z(G), (Z::p() - 1) >> (k + 1));
            for (int i = 1 << (k - 1); i < (1 << k); i++) {
                w[i * 2] = w[i];
                w[i * 2 + 1] = w[i] * u;
            }
            k++;
        }
    }
    for (int k = 1; k < n; k *= 2)
        for (int i = 0; i < n; i += 2 * k)
            for (int j = 0; j < k; j++) {
                Z u = a[i + j], v = a[i + j + k] * w[k + j];
                a[i + j] = u + v;
                a[i + j + k] = u - v;
            }
    if (inv) {
        reverse(a.begin() + 1, a.end());
        Z invn = Z(n).inv();
        for (auto &x : a) x *= invn;
    }
}
template<class Z>
vector<Z> conv(vector<Z> a, vector<Z> b, const int G = 3) {
    if (a.empty() || b.empty()) return {};
    int n = 1, tot = a.size() + b.size() - 1;
    while (n < tot) n <= 1;
    a.resize(n), b.resize(n);
    ntt(a, G), ntt(b, G);
    for (int i = 0; i < n; i++) a[i] = a[i] * b[i];
    ntt(a, G, true);
    a.resize(tot);
    return a;
}

```

10.3 FWT [c53534]

```

template<class T> void fwt(vector<T> &a, bool inv = false) {
    // Or : [[1, 0], [1, 1]] <-> [[1, 0], [-1, 1]]
    // And: [[1, 1], [0, 1]] <-> [[1, -1], [0, 1]]
    // Xor: [[1, 1], [1, -1]] <-> [[0.5, 0.5], [0.5, -0.5]]
    // *inv_2 instead of /2 for Z
    int n = __lg(int(a.size()));
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < 1 << n; j++) {
            if (j >> i & 1) { // And
                T x = a[j ^ (1 << i)];
                T y = a[j];
                if (!inv) {
                    a[j ^ (1 << i)] = x + y;
                    a[j] = y;
                } else {
                    a[j ^ (1 << i)] = x - y;
                    a[j] = y;
                }
            }
        }
    }
}

```

10.4 Polynomial [F07969]

```
template<int P = 998244353, int G = 3>
struct Poly : public vector<Mint<P>> {
    using Z = Mint<P>;
    explicit Poly(int n = 0) : vector<Z>(n) {}
    Poly(const vector<Z> &a) : vector<Z>(a) {}
    Poly(const initializer_list<Z> &a) : vector<Z>(a) {}
    template<class InputIt, class = _RequireInputIter<InputIt>>
    Poly(InputIt first, InputIt last) : vector<Z>(first, last) {}
    Poly operator-() const {
        vector<Z> res(this->size());
        for (int i = 0; i < int(res.size()); i++) res[i] = -(*this)[i];
        return Poly(res);
    }
    friend Poly operator+(Poly a, Poly b) {
        a.resize(max(a.size(), b.size()));
        for (int i = 0; i < b.size(); i++) a[i] += b[i];
        return a;
    }
    friend Poly operator-(Poly a, Poly b) {
        a.resize(max(a.size(), b.size()));
        for (int i = 0; i < b.size(); i++) a[i] -= b[i];
        return a;
    }
    friend Poly operator*(Poly a, Poly b) {
        return conv(a, b, G);
    }
    friend Poly operator*(Poly a, Z x) {
        for (int i = 0; i < a.size(); i++) a[i] *= x;
        return a;
    }
    friend Poly operator/(Poly a, Z x) {
        for (int i = 0; i < a.size(); i++) a[i] /= x;
        return a;
    }
    Poly &operator+=(Poly a) { return *this = *this + a; }
    Poly &operator-=(Poly a) { return *this = *this - a; }
    Poly &operator*=(Poly a) { return *this = *this * a; }
    Poly &operator*=(Z a) { return *this = *this * a; }
    Poly &operator/=(Z a) { return *this = *this / a; }
    Poly shift(int k) const {
        if (k >= 0) {
            auto b = *this;
            b.insert(b.begin(), k, 0);
            return b;
        } else if (this->size() <= -k) {
            return Poly();
        } else {
            return Poly(this->begin() + (-k), this->end());
        }
    }
    Poly trunc(int k) const {
        Poly f = *this; f.resize(k); return f;
    }
    Poly deriv() const {
        if (this->empty()) return Poly();
        Poly res(this->size() - 1);
        for (int i = 0; i < this->size() - 1; i++)
            res[i] = (*this)[i + 1] * (i + 1);
        return res;
    }
    Poly integr() const {
        Poly res(this->size() + 1);
        for (int i = 0; i < this->size(); i++)
            res[i + 1] = (*this)[i] / (i + 1);
        return res;
    }
    Poly inv(int m) const {
        Poly x((*this)[0].inv());
        int k = 1;
        while (k < m) {
            k *= 2;
            x = (x * (Poly{2} - trunc(k) * x)).trunc(k);
        }
        return x.trunc(m);
    }
    Poly log(int m) const {
        return (deriv() * inv(m)).integr().trunc(m);
    }
    Poly pow(ll k, int m) const {
        if (k == 0) { Poly res(m); res[0] = 1; return res; }
        int i = 0;
        while (i < this->size() && (*this)[i] == 0) i++;
        if (i == this->size() || i > 0 && k > (m - 1) / i) return Poly(m);
        Z v = (*this)[i];
        auto f = shift(-i) * v.inv();
        return (f.log(m - i * k) * Z(k)).exp(m - i * k).shift(i * k) * power(v, k);
    }
    Poly sqrt(int m) const { // need QuadraticResidue
        int k = 0;
        while (k < this->size() && (*this)[k] == 0) k++; // 找前導零
        if (k == this->size()) return Poly(m); // 全零多項式
        if (k % 2 != 0) return Poly(); // 無解: 最低次項為奇數
        int s = QuadraticResidue((*this)[k]);
        if (s == -1) return Poly(); // 無解: 係數無平方根
        int offt = k / 2, r = m - offt;

```

```
if (r <= 0) return Poly(m);
Poly h = this->shift(-k) * (*this)[k].inv(), g{1};
for (int i = 1; i < r; i <= 1) {
    int len = i < 1;
    g = (g + h.trunc(len) * g.inv(len)).trunc(len) / 2;
}
g.resize(r);
g = (g * Z(s)).shift(offt).trunc(m);
return g;
}
Poly exp(int m) const {
    Poly x{1};
    int k = 1;
    while (k < m) {
        k *= 2;
        x = (x * (Poly{1} - x.log(k) + trunc(k))).trunc(k);
    }
    return x.trunc(m);
}
Poly mult(Poly b) const {
    if (b.empty()) return Poly();
    int n = b.size();
    reverse(b.begin(), b.end());
    return ((*this) * b).shift(-(n - 1));
}
vector<Z> eval(vector<Z> x) const {
    if (this->size() == 0) return vector<Z>(x.size(), 0);
    const int n = max(x.size(), this->size());
    vector<Poly> q(4 * n);
    vector<Z> ans(x.size());
    x.resize(n);
    function<void(int, int, int)> build = [&](int p, int l, int r) {
        if (r - l == 1) {
            q[p] = Poly{1, -x[l]};
        } else {
            int m = (l + r) / 2;
            build(2 * p, l, m);
            build(2 * p + 1, m, r);
            q[p] = q[2 * p] * q[2 * p + 1];
        }
    };
    build(1, 0, n);
    function<void(int, int, int, const Poly &)> work = [&](int p, int l, int r, const Poly &num) {
        if (r - l == 1) {
            if (l < int(ans.size())) ans[l] = num[0];
        } else {
            int m = (l + r) / 2;
            work(2 * p, l, m, num.mult(q[2 * p + 1]).resize(m - l));
            work(2 * p + 1, m, r, num.mult(q[2 * p]).resize(r - m));
        }
    };
    work(1, 0, n, mult(q[1].inv(n)));
    return ans;
}
};
```

10.5 Barlekamp Massey [853989]

```
// find s[i] = c[0]
// * s[i - 1] + c[1] * s[i - 2] + ... + c[k - 1] * s[i - k]
// return : c(x) = 1 - c_0x - c_1x^2 - ... - c_{k-1}x^k
template<int P = 998244353>
Poly<P> berlekampMassey(const Poly<P> &s) { // 0(n^2)
    Poly<P> c, oldC;
    int f = -1;
    for (int i = 0; i < s.size(); i++) {
        auto delta = s[i];
        for (int j = 1; j <= c.size(); j++)
            delta -= c[j - 1] * s[i - j];
        if (delta == 0) continue;
        if (f == -1) {
            c.resize(i + 1);
            f = i;
        } else {
            auto d = oldC;
            d *= -1;
            d.insert(d.begin(), 1);
            Mint<P> df1 = 0;
            for (int j = 1; j <= d.size(); j++)
                df1 += d[j - 1] * s[f + 1 - j];
            assert(df1 != 0);
            auto coef = delta / df1;
            d *= coef;
            Poly<P> zeros(i - f - 1);
            zeros.insert(zeros.end(), d.begin(), d.end());
            d = zeros;
            auto temp = c;
            c += d;
            if (i - temp.size() > f - oldC.size()) {
                oldC = temp;
                f = i;
            }
        }
    }
    c *= -1;
    c.insert(c.begin(), 1);
    return c;
}
```

10.6 Linear Recurrence [95d930]

```
// a_i = c[0] *
// a_{i-1} + c[1] * a_{i-2} + ... + c[k-1] * a_{i-k}
// q(x) = 1 - c[0] * x - c[1] * x^2 - ... - c[k-1] * x^k
// p(x) = (a * q).trunc(c.size())
// return : a_n
template<int P = 998244353> // O(m log m log n)
Mint<P> linearRecurrence(Poly<P> p, Poly<P> q, ll n) {
    int m = q.size() - 1;
    while (n > 0) {
        auto nq = q;
        for (int i = 1; i <= m; i += 2) nq[i] *= -1;
        auto np = p * nq;
        nq = q * nq;
        for (int i = 0; i < m; i++) p[i] = np[i * 2 + n % 2];
        for (int i = 0; i <= m; i++) q[i] = nq[i * 2];
        n /= 2;
    }
    return p[0] / q[0];
}
```

Prime	Root	Prime	Root
7681	17	167772161	3
12289	11	104857601	3
40961	3	985661441	3
65537	3	998244353	3
786433	10	1107296257	10
5767169	3	2013265921	31
7340033	3	2810183681	11
23068673	3	2885681153	3
469762049	3	605028353	3
2061584302081	7	1945555039024054273	5
2748779069441	3	9223372036737335297	3

11 Else

11.1 Segment Manipulation [30596f]

```
auto era = [&](set<pair<int, int>> &st, int L, int R) {
    auto it = st.lower_bound({L, -1});
    if (it != st.begin() && prev(it)->second >= L) --it;
    while (it != st.end() && it->first <= R) {
        auto [l, r] = *it;
        it = st.erase(it);
        if (l < L) st.emplace(l, L - 1);
        if (R < r) st.emplace(R + 1, r);
    }
};
// remove + 1 for strictly overlap
auto ins = [&](set<pair<int, int>> &st, int L, int R) {
    auto it = st.lower_bound({L, -1});
    if (it != st.begin() && prev(it)->second + 1 >= L) --it;
    while (it != st.end() && it->first <= R + 1) {
        L = min(L, it->first);
        R = max(R, it->second);
        it = st.erase(it);
    }
    st.emplace(L, R);
};
auto find = [&](const set<pair<int, int>> &st, int i) {
    auto it = st.lower_bound({i + 1, -1});
    if (it != st.begin()) {
        --it;
        if (it->second >= i) return it;
    }
    return st.end();
};
```

11.2 Python [7c66a4]

```
from decimal import * # 高精度浮點數
from fractions import * # 分數
from random import *
from math import *
# set decimal prec bigger if it could overflow in precision
setcontext(
    Context(prec=10, Emax=MAX_EMAX, rounding=ROUND_FLOOR))
# read and print
x = int(input())
a, b, c = list(map(Fraction, input().split()))
arr = list(map(Decimal, input().split()))
print(*arr)
# set
st = set(); st.add((a, b)); st.remove((a, b))
if not (a, b) in st:
    # dict
    d = dict(); d[(a, b)] = 1; del d[(a, b)]
    for (a, b) in d.items():
        # random
        arr = [randint(1, r) for i in range(size)]
        choice([8, 6, 4, 1]) # random pick one
        shuffle(arr)
```

11.3 Fraction [62f33d]

```
template<class T> struct Fraction {
    T n, d;
    void reduce() {
        T g = gcd(abs(n), abs(d));
```

```
n /= g, d /= g;
    if (d < 0) n = -n, d = -d;
}
Fraction(T n = 0, T d = 1) : n(n), d(d)
{ assert(d != 0); reduce(); }
Fraction(const string &str) {
    char slash;
    if (str.find('/') != -1) {
        string x = str.substr(0, str.find('/'));
        string y = str.substr(str.find('/') + 1);
        n = stobigint(x), d = stobigint(y);
    } else {
        n = stobigint(str), d = 1;
    }
    Fraction(n, d);
}
Fraction operator+(Fraction rhs) const
{ return Fraction(n * rhs.d + rhs.n * d, d * rhs.d); }
Fraction operator-(Fraction rhs) const
{ return Fraction(n * rhs.d - rhs.n * d, d * rhs.d); }
Fraction operator*(Fraction rhs) const
{ return Fraction(n * rhs.n, d * rhs.d); }
Fraction operator/(Fraction rhs) const {
    assert(rhs.n != 0);
    return Fraction(n * rhs.d, d * rhs.n);
}
friend istream &operator>>(istream &is, Fraction &f) {
    string s; is >> s;
    f = Fraction(s);
    return is;
}
friend ostream &operator<<(ostream &os, const Fraction &f) {
    if (f.d == 1) os << f.n;
    else os << f.n << "/" << f.d;
    return os;
}
bool operator
==(Fraction b) const { return n * b.d == b.n * d; }
bool operator
!=(Fraction b) const { return n * b.d != b.n * d; }
bool operator
<(Fraction b) const { return n * b.d < b.n * d; }
};
```

11.4 Bigint [024a84]

```
struct Bigint { // not support hex division
private:
    static const int digit = 9; // hex: 7
    static const int base = 10; // hex: 16
    static const int B = power(10, digit);
    Bigint(vector<int> x, int sgn) : x(x), sgn(sgn) {}
    template<class U> vector<int> norm(vector<U> a) {
        if (a.empty()) return {0};
        for (int i = 0; i < a.size(); i++) {
            U c = a[i];
            a[i] %= B;
            if (c / B)
                if (i == a.size() - 1) a.push_back(c);
                else a[i + 1] += c;
        }
        while (a.size() > 1 && a.back() == 0) a.pop_back();
        return {a.begin(), a.end()};
    }
    void resign() { sgn = x.back() == 0 ? 1 : sgn; }
    int toInt(char c) const {
        if (isdigit(c)) return c - '0';
        else return c - 'A' + 10;
    }
    char toChar(int c) const {
        if (c < 10) return c + '0';
        else return c - 10 + 'A';
    }
public:
    int sgn = 1;
    vector<int> x; // 反著存
    Bigint() : x {0}, sgn(1) {}
    Bigint(ll a) { *this = Bigint(std::to_string(a)); }
    Bigint(string s) {
        if (s.empty()) { *this = Bigint(); return; }
        if (s[0] == '-') s.erase(s.begin()), sgn = -1;
        int add = 0, cnt = 0, b = 1;
        while (s.size()) {
            if (cnt == digit) x.push_back(add), add = cnt = 0, b = 1;
            add += toInt(s.back()) * b;
            cnt++, b *= base;
            s.pop_back();
        }
        if (add) x.push_back(add);
        x = norm(x);
    }
    int size() const { return x.size(); }
    Bigint abs() const { return Bigint(x, 1); }
    string to_string() const {
        string res;
        for (int i = 0; i < x.size(); i++) {
            string add; int v = x[i];
            for (int j = 0; j < digit; j++)
                add += toChar(v % base), v /= base;
            res += add;
        }
        while (res.size() > 1 && res.back() == '0') res.pop_back();
```

```

    if (sgn == -1) res += '-';
    reverse(res.begin(), res.end());
    return res;
}
Bigint operator-() const { return Bigint(x, -sgn); }
friend Bigint operator+(Bigint a, Bigint b) {
    if (a.sgn != b.sgn) return a - (-b);
    int n = max(a.size(), b.size());
    a.x.resize(n), b.x.resize(n);
    for (int i = 0; i < n; i++) a.x[i] += b.x[i];
    a.x = a.norm(a.x), a.resign();
    return a;
}
friend Bigint operator-(Bigint a, Bigint b) {
    if (a.sgn != b.sgn) return a + (-b);
    if (a.abs() < b.abs()) return -(b - a);
    int n = max(a.size(), b.size());
    a.x.resize(n), b.x.resize(n);
    for (int i = 0; i < n; i++) {
        a.x[i] -= b.x[i];
        if (a.x[i] < 0) a.x[i] += B, a.x[i + 1]--;
    }
    a.x = a.norm(a.x), a.resign();
    return a;
}
friend istream &operator>>(istream &is, Bigint &a)
{ string v; is >> v; a = Bigint(v); return is; }
friend ostream &operator<<(ostream &os, Bigint a)
{ os << a.to_string(); return os; }
friend bool operator<(Bigint a, Bigint b) {
    if (a.sgn != b.sgn) return a.sgn < b.sgn;
    if (a.x.size() != b.x.size())
        return a.x.size() < b.x.size();
    else for (int i = a.x.size() - 1; i >= 0; i--)
        if (a.x[i] != b.x[i]) return a.x[i] < b.x[i];
    return 0;
}
friend bool operator>(Bigint a, Bigint b) {
    if (a.sgn != b.sgn) return a.sgn > b.sgn;
    if (a.x.size() != b.x.size())
        return a.x.size() > b.x.size();
    else for (int i = a.x.size() - 1; i >= 0; i--)
        if (a.x[i] != b.x[i]) return a.x[i] > b.x[i];
    return 0;
}
friend bool operator==(const Bigint &a, const Bigint &b)
{ return a.sgn == b.sgn && a.x == b.x; }
friend bool operator!=(const Bigint &a, const Bigint &b)
{ return a.sgn != b.sgn || a.x != b.x; }
friend bool operator>=(const Bigint &a, const Bigint &b)
{ return a == b || a > b; }
friend bool operator<=(const Bigint &a, const Bigint &b)
{ return a == b || a < b; }
};
Bigint abs(const Bigint &a) { return a.abs(); }
Bigint stoBigint(const string &s) { return Bigint(s); }

```

11.5 Multiple [9011f6]

```

// Require: Mint, NTT
using i128 = __int128_t;
const int P1 = 167772161;
const int P2 = 104857601;
const int P3 = 998244353;
const i128 T1 = i128(P2) * P3;
const i128 T2 = i128(P1) * P3;
const i128 T3 = i128(P1) * P2;
const int I1 = Mint<P1>(T1).inv().x;
const int I2 = Mint<P2>(T2).inv().x;
const int I3 = Mint<P3>(T3).inv().x;
const i128 M = i128(P1) * P2 * P3;
vector<i128> arbitraryMult
(const vector<int> &a, const vector<int> &b) {
    int n = a.size(), m = b.size();
    auto x = conv(vector<Mint<P1>>(a.begin(), a.end()), vector<Mint<P1>>(b.begin(), b.end()));
    auto y = conv(vector<Mint<P2>>(a.begin(), a.end()), vector<Mint<P2>>(b.begin(), b.end()));
    auto z = conv(vector<Mint<P3>>(a.begin(), a.end()), vector<Mint<P3>>(b.begin(), b.end()));
    vector<i128> res(x.size());
    for (int i = 0; i < x.size(); i++)
        res[i] = (x[i].x * T1 % M * I1 +
                  y[i].x * T2 % M * I2 +
                  z[i].x * T3 % M * I3) % M;
    return res;
}
public:
friend Bigint operator*(Bigint a, const Bigint &b) {
    a.x = a.norm(arbitraryMult(a.x, b.x));
    a.sgn *= b.sgn, a.resign();
    return a;
}

```

11.6 Division [5be5f1]

```

private:
pair<vector<int>, int> smallDivMod(vector<int> a, int v) {
    ll add = 0;
    for (int i = a.size() - 1; i >= 0; i--) {
        add = add * B + a[i];
        int q = add / v;
    }
}

```

```

    a[i] = q, add %= v;
}
return {norm(a), add};
}
friend Bigint operator<<(Bigint a, int k) {
    if (!a.x.empty()) {
        vector<int> add(k, 0);
        a.x.insert(a.x.begin(), add.begin(), add.end());
    }
    return a;
}
friend Bigint operator>>(Bigint a, int k) {
    a.x = vector<int>
        >(a.x.begin() + min(k, int(a.x.size())), a.x.end());
    a.x = a.norm(a.x);
    return a;
}
public:
friend pair<Bigint
    , Bigint> divmod(const Bigint &a, const Bigint &b) {
    int qsgn = a.sgn * b.sgn, rsgn = a.sgn;
    Bigint av = a.abs(), bv = b.abs();
    Bigint quot, rem;
    if (av < bv) {
        rem = av;
    } else if (bv.size() == 1) {
        auto [qd, r] = quot.smallDivMod(av.x, bv.x[0]);
        quot.x = qd;
        rem = Bigint((ll)r);
    } else {
        Bigint inv = 1LL * B * B / bv.x.back();
        Bigint pre = 0, res = 0;
        int d = av.size() + 1 - bv.size();
        int cur = 2, bcur = 1;
        while (inv != pre || bcur < bv.size()) {
            bcur = min(bcur < 1, bv.size());
            res.x = {bv.x.end() - bcur, bv.x.end()};
            pre = inv;
            inv = inv
                * ((Bigint(2) << (cur + bcur - 1)) - inv * res);
            cur = min(cur < 1, d);
            inv.x = {inv.x.end() - cur, inv.x.end()};
        }
        inv.x = {inv.x.end() - d, inv.x.end()};
        res = (av * inv) >> av.size();
        Bigint mul = res * bv;
        while (mul + bv <= av) res = res + 1, mul = mul + bv;
        quot.x = quot.norm(res.x);
        rem = av - mul; // 重用 mul, 不再重算一次 quot * b
    }
    quot.sgn = qsgn, quot.resign();
    rem.sgn = rsgn, rem.resign();
    return {quot, rem};
}
friend Bigint operator/(const Bigint
    &a, const Bigint &b) { return divmod(a, b).first; }
friend Bigint operator%(const Bigint
    &a, const Bigint &b) { return divmod(a, b).second; }
Bigint gcd(Bigint a, Bigint b) {
    while (b != 0) {
        Bigint r = a % b;
        a = b, b = r;
    }
    return a;
}
}

```

11.7 Division-Python [110bd8]

```

from decimal import * # 無誤差浮點數
setcontext(
    Context(prec=4000000, Emax=4000000, rounding=ROUND_FLOOR))
t = int(input())
for i in range(t):
    a, b = map(Decimal, input().split())
    d, m = divmod(a, b)
    print(d, m)

```